

Product Engineering Process (PEP) Directive

设计控制程序(PEP)

SSME

XPQR 4.4/01

55 21 906 - AND - 708 - 16

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Restricted

Siemens Shanghai Medical Equipment Ltd.

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History/修改历史

Nr.	Page	Version	Change description	CR No.
1	All	01	Initial creation, the content transferred from 5521906-AND-017-005 due to SSME QMS restructuring. This version is also with consideration of P00969 from SSME QMS system.	n.a.
2	All	02	0. Based on CAPA management review(XP-201504) result: Add check item in main review R4/R5.2 template for USA market documents 1. Correct the error in reference document 2. Align with HQ XP PEP for spare parts plan in M400, inform region in EOS part. 3. Add usability engineer file in the related milestone based on the suggestion from Q 4. Add criteria for a project implemented by Project or ECO in 6.5.2, suggested from Q 5. Add TPSW related file in the main review template(R2, R3, R4, R5.2) 6. Add Reused components verification evaluation report in R2 template 7. Add Technical data from R&D in R4 template 8. Add Acceptance Checklist from SCM in design transfer protocol template	PCR 10293
3	All	03	Integrate requirements from QR 24 PLM Core Process 1. Add M270 for CUT 2. Change original R5.0 to R-CUT, R5.2 to R5 3. Add Related document in the R3 template 4. Add M100/M500 to milestone list 5. Remove main content for M400/M450, but refer to [12] 6. Add M450/M500 decision protocol template 7. Change “MRS” -> “PRS”, change “TR” to “PR” 8. Charm SP_00221083, add reuse evaluation for R1 template as draft version 9. Chapter 5.5 is updated (both content and structure), add SSIT as example for R3 10. Add 5521906-AND-285 Clinical Evaluation in European Union> as a reference documentation and make clinical performance assessment is mandatory before M150	PCR 10377
4	All	04	Change Content: - Defect 91356: figure 1, M400 is marked at wrong place. - Add QA as individual reviewer during review (6.5.4), the sentence aligned with GGXP PEP. - Update R4 check list (add ST spec) - Update design transfer protocol based on new design transfer WI content	PCR 10502
5	All	05	E-QMS project	PCR 10626
6	All	06	Change Content: Implement EU MDR requirements.	PCR 10643

			- Defect 151498, main review protocol template for R1, R2, R3, R4, and R5 are updated. The action plan for deviations and restrictions is added into template. - Defect 157485, main review protocol template for R1, R2, R3, R4, and R5 are updated. Add definition of deviation and restriction into template. - Defect 131566, add “Product Related Environmental Aspects and Improvement Targets” as one of the content of R1. - Add spare parts list and IVK list as one of the content of R3 which is the design output for supplier preparation and DHHS preparation.	
7	All	07	Change Content: - Make Post-market Clinical Follow-up Plan as a separated document. Update review template for R5. - Add PSS requirement into main review protocol and update template for R1, R2, R3, R4 and R5.	PCR 10671
8	Appendix	08	Change Content: - Defect 182638, add DMR template as appendix. - Defect 182639, add “Risk Assessment of New Features” into R2 template. - Add “Customer user test report” and “Customer user test assessment” into R5 template. - Add “Usability Assessment of New User Messages” into R3 and R4 template. - Update appendix 3, add dangerous goods list in design transfer protocol.	PCR 10696
9	Appendix	09	Change Content: - Defect 191597, move “Risk Management Report” from R4 template to R5 template. - Defect 191586, add check item of risk assessment for technical master file from internal supplier in R3 template. - Defect 209550, add check item of uploading UDI information into database in R5 template.	PCR 10770
10	Appendix	10	Change Content: - Defect 246090: - Remove DHF/DMR section index number in main review templates. - Update appendix 5, add new documents. - Remove assessment for complaints of predecessor products in R2 template. - Update wording in R2 template, risk assessment to new requirements instead of new features.	PCR 10818
11	Reference& Appendix	11	Change Content: - Defect 271391, modify R5 protocol template, item 58 “US approvals” to “US approvals including device listing”. - Add penetration test plan and report into R4 protocol template. - Add reference document, AI Model Training and Validation Process	PCR 10890

12	All	12	Change Content: - Add product reliability relevant content from HQT QR0 & GD65. - Add requirements of entrusting design & development control - Update Appendix	PCR 10972
13	Page 7, 46, 47,48 Reference& Appendix	13	Change Content: - Refine the responsibility of PSG and define the template of PSG meeting minutes as appendix 14. - Update document ID for [14] - Update template for Production Review Protocol、M400 Protocol、M450 Protocol and M500 Protocol	PCR 11029
14	Page 17, 20, 23, 24, 38	14	Change Content: - Add early procurement & supplier involvement description in Chapter 5.6 & 6.2. - Add main review R2' and M230 description in Chapter 5.5 & 5.6.	PCR 11127
15	Appendix	15	Chang Content: - Update Appendix2 to add instruction of archiving EK slides. - Update Appendix11 to add instruction of archiving EK slides. - Update Appendix12 to add instruction of archiving EK slides. - Update Appendix15 to add instruction of archiving EK slides and update the role name.	PCR 11184
16	All	16	Change Content: - Harmonization with PEP process of XP FO, main modification in Chapter 5&6, incl. definition, wording, description, and content structure. - Update Appendix6, add item11 & 12, process harmonization with XP FO. - Update Appendix7, add item22, process harmonization with XP FO. - Update Appendix7, add item23, fulfill AI algorithm development process. - Update Appendix8, add item29, fulfill AI algorithm development process.	PCR 11369

1. Purpose and scope/目的和适用范围

This QR shall be effective for SSME XP where the QMM [6] is applicable.

It regulates the procedure for planning, developing, supporting and further development of systems, sub-systems, components and services and purchases included therein. It describes the general specifications and typical requirements, which must be complied within all development projects.

Depending on the individual project circumstances, the specifications can be tailored to the requirements, provided a reason is given and the basic concept of this XP QR is not compromised (tailoring). In the case of further developments, for example, the formal M120 and M150 decisions may be omitted, as long as all the required design input documents are available. The reason must be documented in the project-specific QMP.

For software development, [3] is also valid.

本程序适用于在质量管理手册[6]适用范围内的 SSME XP。

本程序规定了对系统、子系统、部件进行计划、技术开发, 技术支持及进一步开发的流程, 维修和采购也包含在内。本程序还描述了所有技术开发项目必须遵守的总方针以及典型要求。

只要不违背质量保证程序基本原则, 本规定可以根据项目的特殊情况进行修改以适合具体的需求(量体裁衣 – 在质量管理计划中描述)。比如, 对于进一步技术开发, 只要提交所有所需的设计输入文件, 可以省略正式 M120 和 M150 决定。具体的原因必须在项目的质量管理计划中描述。

对于软件开发, [3]也适用。

2. Reference document/参考文件

[1]	5521906-AND-705	Change Control Procedure	变更控制程序
[2]	4791600-QMS-700	Supplier Quality Management	供应商质量管理
[3]	5521906-AND-749	Software Development Directive	软件开发指导
[4]	5521906-AND-719	Process Validation and Verification Procedure	过程确认与验证程序
[5]	5521906-AND-748	Customer Complaint Handling Procedure	用户抱怨处理程序
[6]	5521906-AND-700	Quality Management Manual (QMM)	质量管理手册
[7]	5521906-AND-210	Document Management Procedure	文件管理程序
[8]	5521906-AND-752	Product Risk management Procedure	产品风险管理程序
[9]	5521906-AND-702	Management responsibility	管理职责
[10]	5521922-AND-777	Customer Use Test, Clinical Investigation and Monitoring Phase working Instruction	客户应用测试、临床试验及产品质量监控作业指导书
[11]	5521922-AND-723	Customer Documentation Working Instruction	用户文档管理作业指导书
[12]	5521906-AND-760	Product Phase-out Process at Product Lifecycle Management Procedure	产品生命周期管理中的产品逐步退出流程
[13]	5521906-AND-778	Clinical Evaluation and Clinical Investigation in European Union	欧盟临床评估和临床研究
[14]	5521906-AND-711	Product Related EHS Protection Operating Procedure	与产品相关的 EHS 管理程序

[15] 5521906-AND-706	Product and Solution Security Directive	产品和解决方案（信息）安全程序
[16] 5521922-AND-726	AI Model Training and Validation Process	AI 模型训练与验证流程
[17] 11106079-AND-02S	QR 28 - Product Reliability Management	QR 28 – 产品可靠性管理
[18] 5521906-AND-728	Corrective and Preventive Actions (CAPA) Procedure	纠正和预防措施程序
[19] 4826223-AND-13E	Definitions	定义
[20] 5521922-AND-717	Third-party Software Working Instruction	第三方软件作业指导书
[21] 5521906-AND-774	Product Usability Management Procedure	产品可用性管理程序
[22] 4826223-AND-10E	Product Engineering Process (XP FO)	设计控制程序（XP FO）

3. Abbreviations and Definitions/缩略语与定义

3.1 Abbreviations/缩略语

CE	European Conformity (Conformité Européene)	欧盟符合性认证
CS	Customer Service	客户服务
DHR	Device History Record	产品追溯性记录
DMR	Device Master Record	产品制造性文档
DHF	Design History File	开发历史文档
EMC	Electro Magnetic Compatibility	电磁兼容性
FMEA	Failure Modes and Effects Analysis	失效模式和效果分析
FS	Functional Specification	功能规范
FPP	Financial product plan	产品财务计划
IVK	Installed Volume Component	须跟踪的现场安装部件
M120-M500	Main Decision	主要决定
MDD	Medical Device Directive	医疗器材指令
EU MDR	EU Medical Device Regulation	欧盟医疗器材法规
MQR	Medical Quality Regulation	医疗器械质量法规
PRS	Product Requirement Specification	产品需求规范
R&D	SSME XP Research and Development	SSME XP 研究与开发部门
PPM	Product portfolio management	产品组合管理
PMP	Project Management Plan	项目管理计划
PSG	Project Steering Group	项目领导小组
SFS	System Functional Specification, can also be	系统功能规范，也可以命名为

	called SRS (System Requirement Specification)	系统需求规范
SSFS	Subsystem Functional Specification, can also be called SSRS (Subsystem Requirement Specification)	子系统功能规范，也可以命名为子系统需求规范
SWFS	Software Function Specification, can also be called SWRS (Software Requirement Specification)	软件功能规范，也可以命名为软件需求规范
QMP	Quality Management Plan	质量管理计划
QT	Quality Technology	质量部
R1-R5	Main Reviews	主要评审
RCT	Review clinical trial / clinical investigation	针对“临床试验”的评审
RCUT	Review of Customer Use Test	针对“客户应用测试”的评审
SW	Software	软件
PR1-PR5	Partial Review	部分评审
UL	Underwriters' Laboratories	(美国)保险商实验室
WPP/GWB	Economic Product Plan/ Economic Value-Added	经营产品计划/经营增值

3.2 Definitions/定义

3.2.1 Engineering sample / 工程样机

A sample based on preliminary development documents (prior to design transfer), often with a defined, reduced set of requirements which is authorized by the originator. See definition in [19]

根据初步设计文件(在设计转移之前)制造的样机。通常只会实现部分需求。参考在[19]中的定义

3.2.2 Functional sample / 功能样机

A sample based on approved development documents (insofar as it is relevant for a functional sample) – preferably from production or from a supplier (orders can be placed informally) – which is authorized by the originator. See definition in [19]

根据已批准的设计文件(仅适用于功能样机)制造的样机。建议通过量产方式或供应商生产制造(可以通过非正式订单)。参考在[19]中的定义

3.2.3 Production sample / 生产样机

A sample which is fully functional, produced on the basis of approved production documents and which has serial production status (including bought-in parts from suppliers). See definition in [19]

根据已批准的生产文件制造的样机。必须实现全部功能，并且与量产状态一致(包括从供应商采购的部件)。参考在[19]中的定义

4. Flow chart/流程图

n.a.

无

5. Process & Requirement/过程和需求

5.1 V-model/Requirement Tracing/ V 型模式/需求跟踪

This QR describes the product engineering process based on the V-model of product engineering.

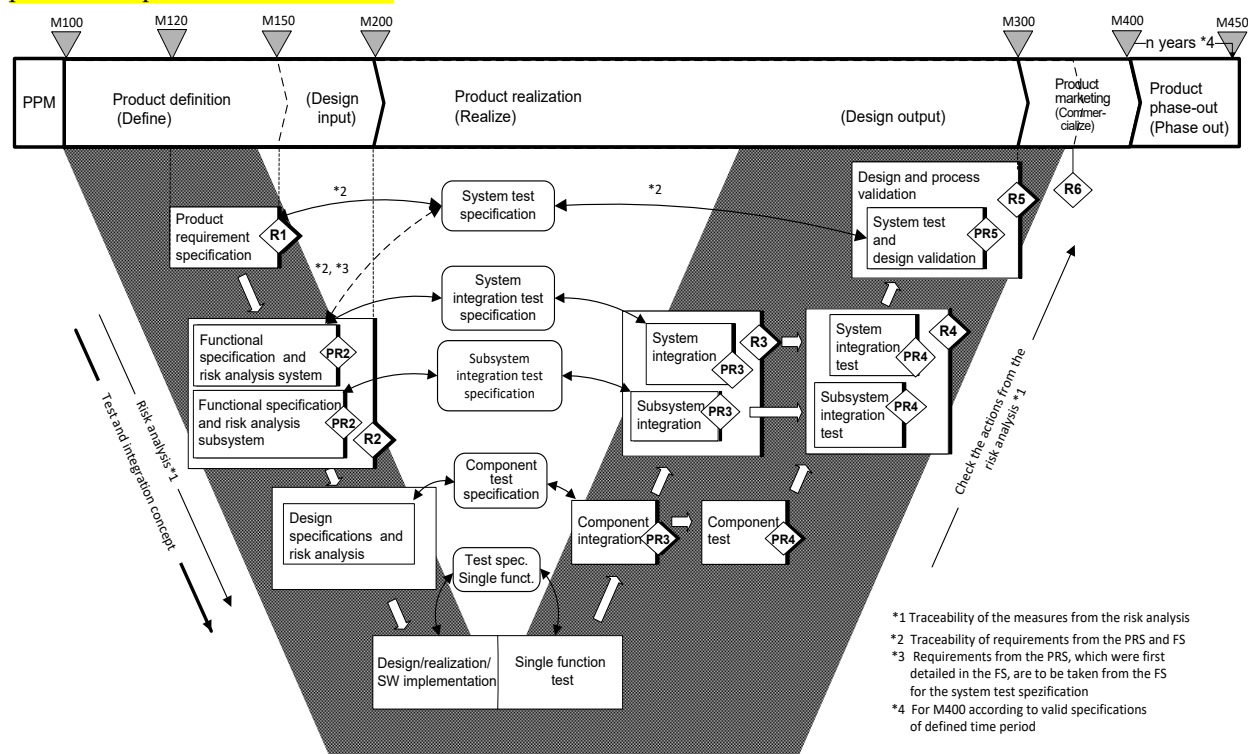
The description of product engineering process in the following chapters shows the complete procedure from the systems perspective. Sub-systems / components similarly follow the V-model (refer to chapter 5.2).

For sub-systems which are sold as independent products require a M300 decision.

本质量保证程序根据产品设计 V 字型模式说明产品设计过程。

本文以下章节所阐述的产品设计的完整步骤是针对系统的。对于子系统及部件，同样也要遵循这一 V 字形模式。(参照 5.2 节)

当子系统作为单独产品销售的情况下，要求作 M300 决定。



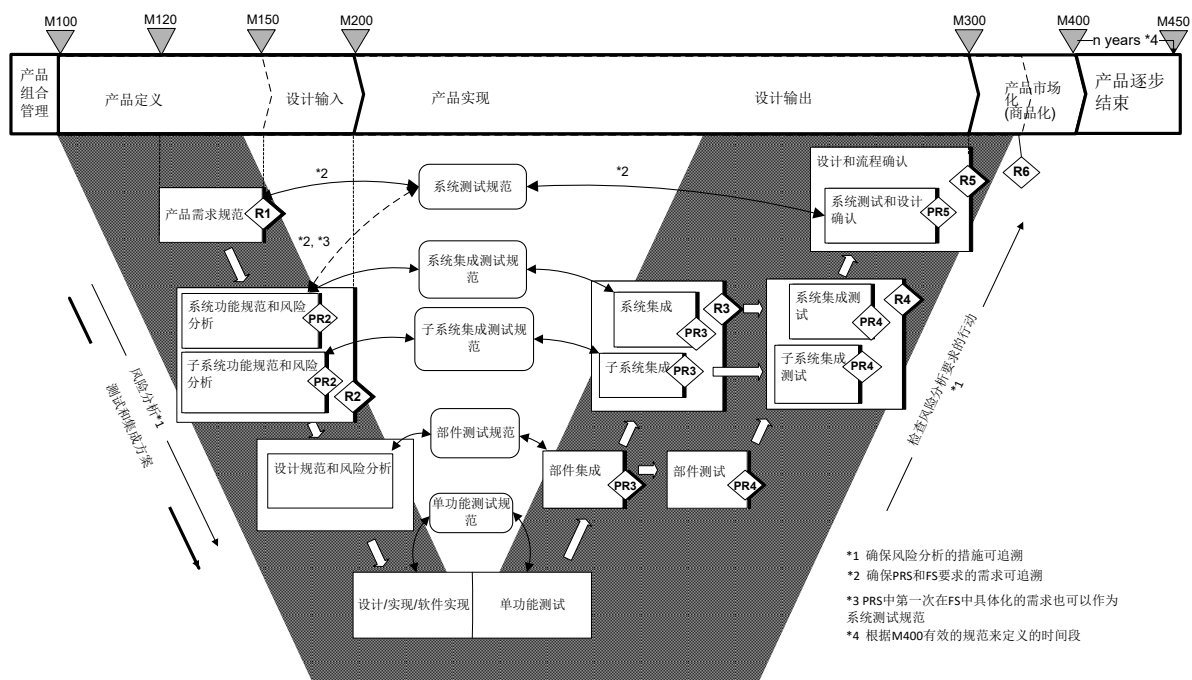


Fig.1/图 1: V-model/V 字型模式

The **Left branch of the V-model** illustrates how requirements are refined from the product requirement specification through to the functional specifications including the corresponding risk analysis through to the design specifications **and implemented in a corresponding design**. In order to trace this process, the relevant documents are brought to a solution. Main reviews R1 and R2 ensure that the requirements are sufficiently and clearly formulated, **that they do not contradict themselves and that they have been fully and properly implemented in the form of verifiable specifications.**

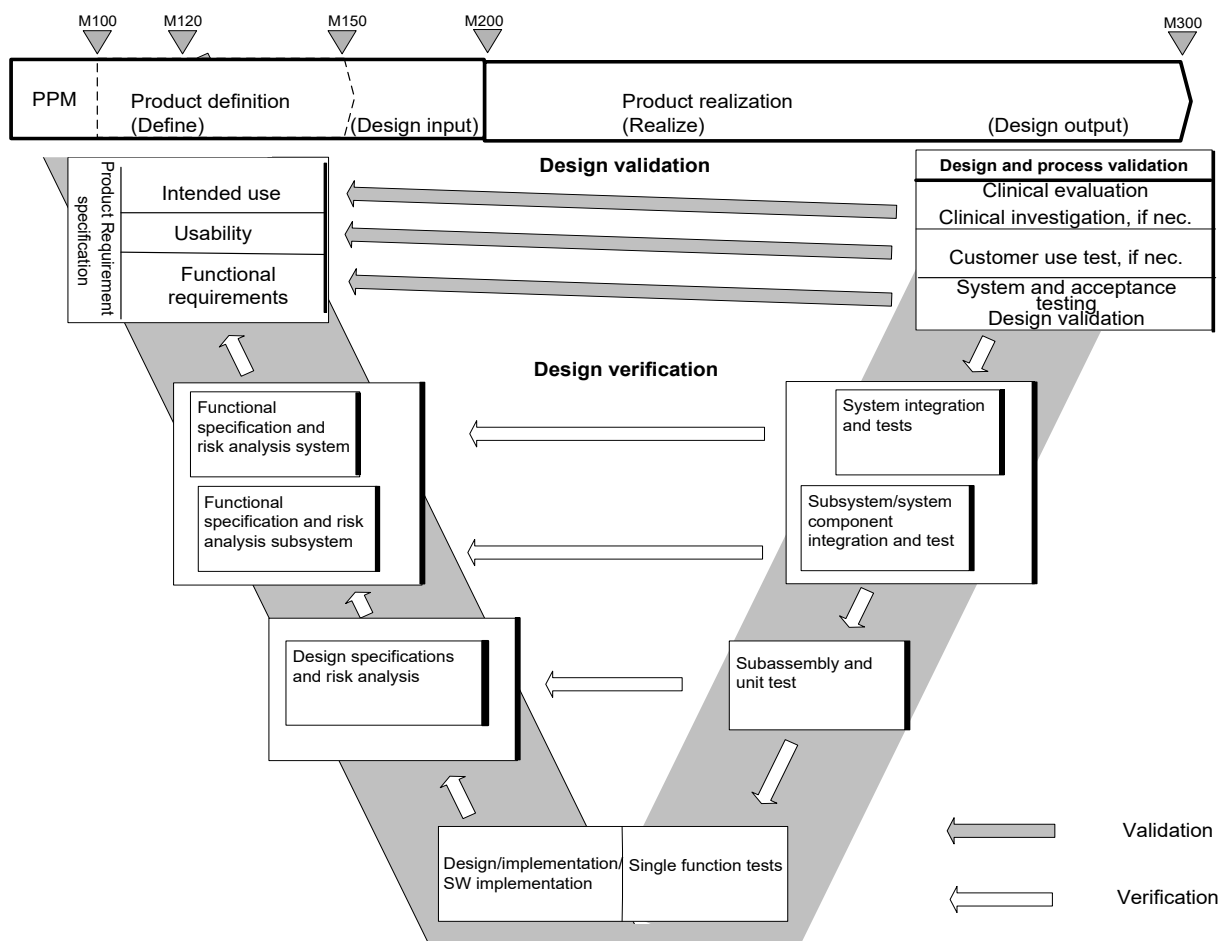
The **right branch of the V-model** shows the requirements for the relevant integration and test phases. The documents to be created and the tests may be tailored to the particular technology (mechanical, electromechanical, electronic, SW). Main reviews R3 to R5 check whether the design and development outputs and the production results meet the requirements and whether the validation has been successfully performed. Test cases are used in the corresponding test phases to verify that the requirements from the product requirement specification, functional specification and the design

V 字型模式左侧表明在产品需求规范中提出的需求如何通过功能规范包括风险分析细化成设计，直至形成设计规范，**并且通过对应的设计实现这些需求**。为了追溯整个过程，需编写相应的文件、归类、分配专门代码并阐明如何实现这些需求的解决方案。主审评 R1 和 R2 阶段的工作是为了确保需求已充分、明确地表达清楚。**而且一个可验证的解决方案可以完全、合理地实现这些需求。**

V 字型模式右侧表明与需求相对应的集成和测试阶段。文件和测试可以根据具体技术分类(机械、机电、电子、软件)进行定制。主审评 R3 到 R5 阶段主要审核设计、开发和生产输出是否满足需求并成功地完成了确认。在测试阶段，测试用例用来验证产品需求规范、功能规范和设计规范提出的需求。必须证明所有的需求都被完全地实现、验证和确认过。**主审评 R6 主要确认产品导入市场的结果。**

specifications have been met. Proof must be provided that all requirements have been fully implemented, verified, and validated. **Main review R6 confirms the conclusion of the product introduction.**

5.2 Design verification and validation/设计验证和确认



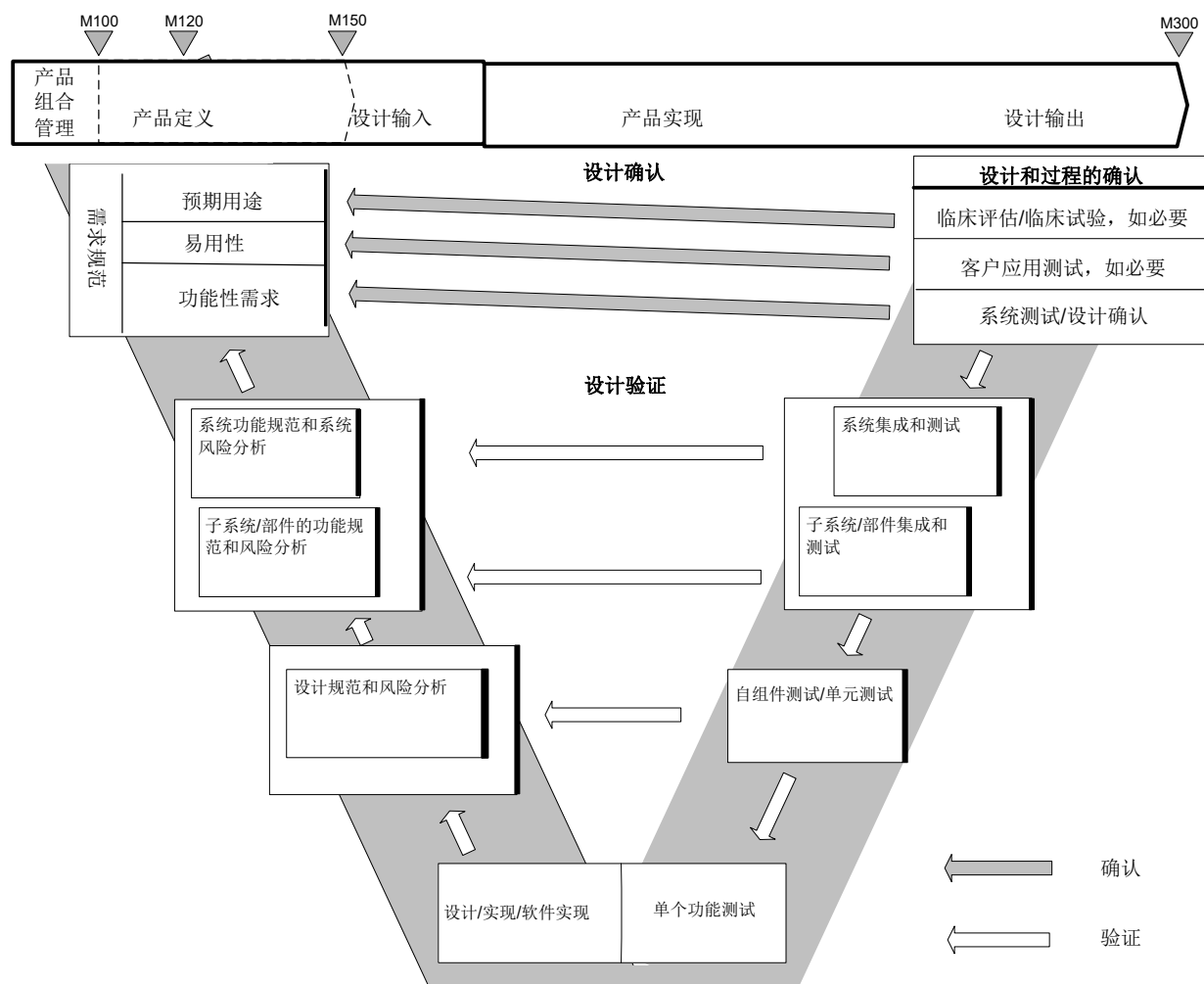


Fig. 2/图 2: Design verification and validation in the V-model/V 字型模式中的设计验证和确认

Design verification

As part of design verification, valid, and objective evidence is used to confirm and document that work results correspond to the specifications or are suitable for the implementation of requirements (“The product was developed correctly”).

Design validation

At the end of the product engineering process, valid and objective evidence is used as part of design validation to confirm (assess) and document that the product specifications were realized and the planned usability as well as user functionality were met for the intended use (“The correct product was developed”).

设计验证

相对应的需求和规范验证结果必须以文件形式写下来，以保证产品开发的正确性。

设计确认

在设计控制程序的最后阶段，相应的设计确认结果必须以文件形式写下来，以保证产品规范、易用性需求以及用户功能真正得到实现和达到预期目标。

5.3 Structure of the Product Engineering Process/产品设计过程结构

The product engineering process in the SSME XP is realized through the product lifecycle management process (PLM process) (see Fig. 3). The product lifecycle begins with product portfolio management (PPM), continues with the precise definition of the product. Its realization and its marketing and ends with product discontinuation (phase out).

SSME XP 的产品设计过程贯穿产品生命周期管理流程 (PLM 流程) (参照图 3)。产品的生命周期从产品组合管理 (PPM) 开始, 在经历产品定义、实现和市场化阶段之后, 中止与产品逐步结束阶段。

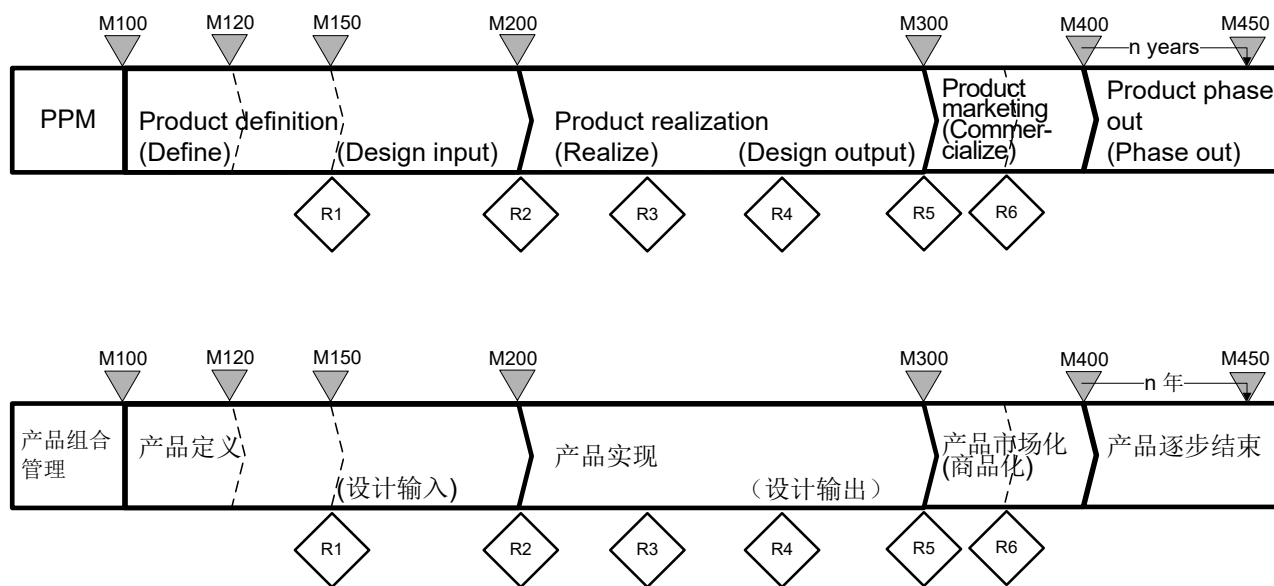


Fig. 3/图 3: Structure of the product engineering process/ 产品设计过程结构

5.3.1 Description of the process phase/过程阶段的说明

In the "Product portfolio management" (PPM) process section, product ideas are collected, evaluated and filtered.

在“产品组合管理” (PPM) 阶段, 需要收集、评估和过滤产品相关的创意。

The "Product definition" process section is divided into the three subphases: Analyze product-specific market, Define marketing and product requirements, and Define realization concept.

产品定义过程分成 3 个子过程: 分析产品对应的市场、定义市场和产品需求、定义实现设想。

The "Product realization" process section includes the start of production.

产品实现过程包括开始生产。

In the "Product marketing" process section, product quality is monitored. Measures are derived from this; the implementation of those measures is tracked and their effectiveness determined.

在产品市场化过程中, 需要监控产品的质量, 并采取相应的措施。这些措施需要跟踪以保证有效性。

The "Product discontinuation (phase out)" process section includes the decision for end of delivery.

产品逐步结束过程包括产品结束发货的决定。

From the "Product definition – Define marketing and product requirements" process section onwards, all process sections are accompanied by the activities of project/quality management (see Section 6.5).

从“产品定义-定义市场和产品需求”过程部分开始, 所有过程阶段均伴随着项目/质量管理活动(参照 6.5 节)。

The SSME XP management provides the resources required for the *process section* through the respective decision (M100 – M300). The decision is based on the documents listed under the heading “Basis for decision”. The results (quality-relevant documents or units), which are in turn the basis for the work packages of the subsequent process section, must be checked by the roles specified under participants or be subjected to a review.

Product engineering generally takes place in the form of a project using project management methods. The project manager initiates project processing in the corresponding XP business processes, which assume responsibility for implementation according to their defined tasks and objectives. The project manager tracks the processing of the project-specific work packages in the corresponding business processes.

It may be necessary to repeat previous process sections or parts thereof following changes or error corrections.

对于各决定阶段(M100-M300)，SSME XP 管理经营部门提供过程阶段需要的资源。根据决定依据标题下所列的文件作出决定。这些结果(质量-相关文档或单元)，反过来成为以后过程阶段分配任务的依据，必须经过测试或者由称为角色的参与者进行评审。

产品设计通常采用项目管理方法，以项目的形式进行。项目经理启动项目，根据相应的职责定义相应的任务和目标，并且跟踪所有过程阶段期间项目相关任务的进展情况。

如果发生更改或纠正，可能需要回到先前的过程阶段，或者其中的部分。

5.3.2 Review items & milestones in PLM/产品生命周期管理中的审核和里程碑

The review items and milestones in the PLM are shown in the figure below.

产品生命周期管理中的审核和里程碑如下图所示。

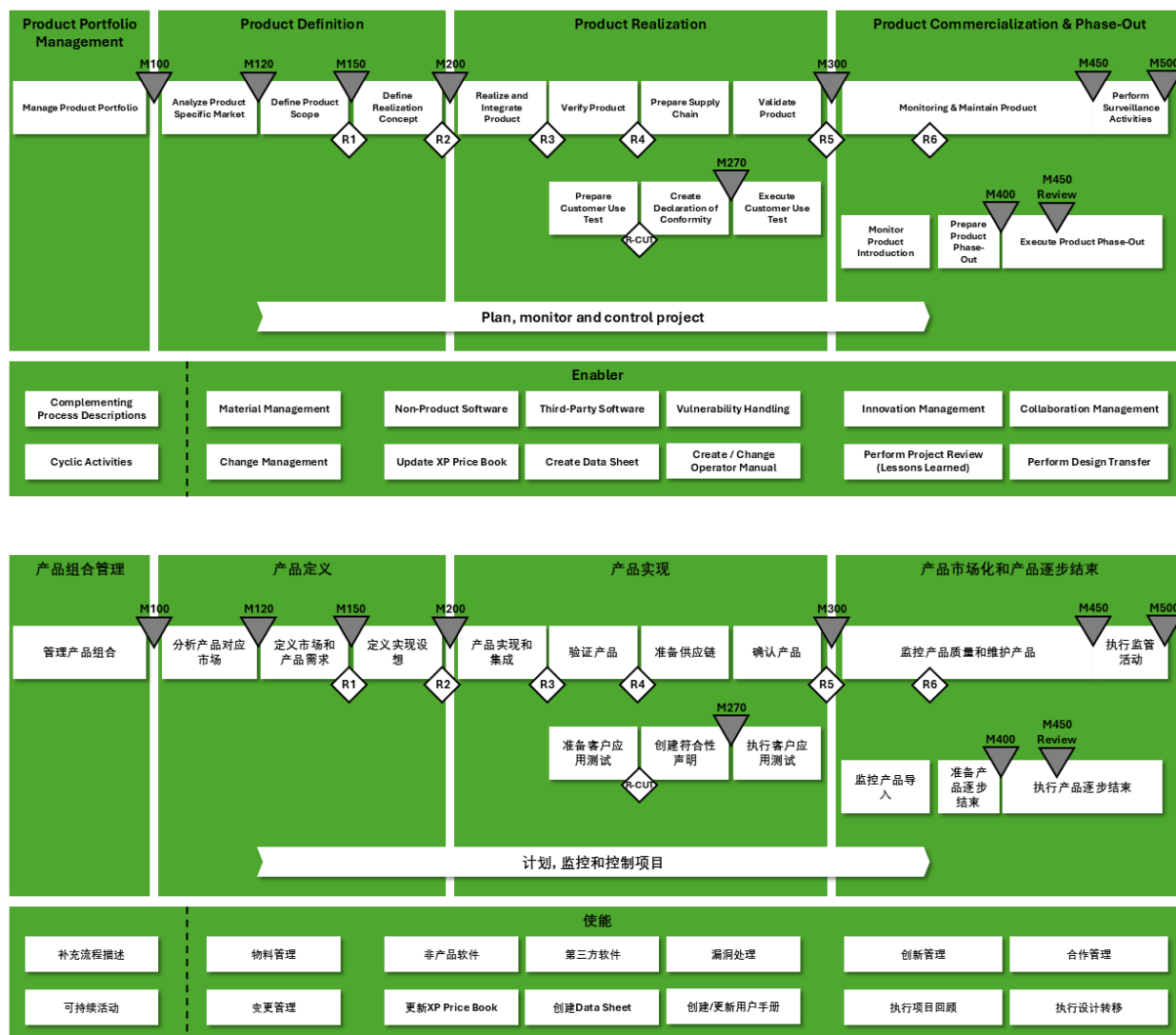


Fig. 4/图 4: Review items and milestones in PLM / 产品生命周期管理中的审评和里程碑

The milestones in the PLM are defined as follows:

- M100: Decision to start the process section product definition (optional, no EK meeting)
- M120: Decision to initiate the product definition
- M150: Decision to commit the realization concept phase
- M200: Decision to commit the design and development
- M270: Product release for customer use test (by Management not by PSG, no EK meeting, presentation and decision in BIT meeting)

定义如下产品生命周期管理中的里程碑:

- M100: 决定启动产品定义流程 (可选, 不需要 EK 会议)
- M120: 决定开始产品定义
- M150: 决定启动产品实现方案阶段
- M200: 决定确认设计和开发阶段
- M270: 决定发运产品进行客户应用测试 (在 BIT 由管理层决定而不是 PSG, 一般不需 EK 会议)

- M300: Delivery release (series)
- M400: Decision about end of delivery (EOD)
- M450 review: Decision about final M450 date
- M450 (EOS): End of the guaranteed spare parts supply period (end of product support)
- M500: End of product lifecycle, detail please refer to [12]

Note:

Normally “Customer Use Test (CUT)” will be implemented after successful Main review R-CUT and M270, detailed process is described in [10].

If a development project does not need a customer use test, a tailoring should be possible. For tailoring a rational shall be prepared and must be accepted within the BIT or M200 - EK meeting.

The individual main reviews have the following applicability with respect to the system, sub-system, and component level:

R1: Requirements for the unit from the point of view of “user”/ “ordering party”. To confirm that the product requirements are adequately defined.

R2: Implementation of the requirements for the unit from the point of view of the "contractor". To confirm that the product realization concept is adequately defined.

R3: Confirm completion of design and development

R4: Confirm completion of product verification activities

R-CUT: Review for customer use test

R5: Review for series release (process validation) (To verify that product validation testing has been completed and the product is ready to be released for general availability)

R6: To confirm and accept the quality key performance indicator

A main review done on the sub-system/component level is a partial review (PR) on the system level.

- M300 : 产品发运的决定
- M400: 停止发运的决定(EOD)
- M450 review : 决定最终 M450 的日期
- M450 (EOS): 决定结束配件供应的日期（结束产品的支持）
- M500: 决定终止产品生命周期, 具体请参考 [12]

注:

通常情况下，只有成功通过了主评审 R-CUT 和 M270 才能开始“客户应用测试”。具体的流程描述见 [10]。

如果在项目中需要裁剪 CUT，必须有充分的理由并在 BIT 或 M200-EK 会议上决定接受。是基于一个全面的开发流程准备和进行的，但会针对有限的客户目标集中在必要的活动上。

对系统、子系统和部件的各个主评审适用下述情况:

R1: 从“使用者”/“用户”角度对机器提出需求。来保证产品被充分的定义。

R2: 从“供货商”角度对单元提出的需求进行实施。确保产品实现方案被充分定义。

R3: 确保设计和开发的完成。

R4: 确保产品验证活动的完成。

R-CUT: 为开始客户应用测试进行审评

R5: 为产品发布进行评审（流程确认）（确保产品确认测试的完成以及确保产品发布的准备已完成）

R6: 确认和接受产品质量性能指标的完成情况

子系统/部件完成的主评审作为整个系统的部分评审(PR)。

5.4 General requirements valid for all process phases/对所有过程阶段适用的一般需求

Concurrent Engineering/同步设计

In the framework of concurrent engineering, it may be necessary to begin a process phase before the previous phase has been formally concluded. This procedure shall be determined in the QMP or a specific guideline. A main review may not be concluded before the previous main review has been formally concluded.

在同步设计工作框架中，可能会有必要在前一过程阶段正式结束之前就开始下一过程阶段。这种程序应当在质量管理计划或专门指导书中确定。但是，如果前一项主评审没有正式结束，则下一项主评审不能结束。

A release stream (as part of a solution stream) is responsible for the release of a medical device and starts the M150. It thus represents a development project in accordance with the quality regulations.

If a development project wants to move the main review R2 closer to R3, below procedures must be followed:

- additional main review R2' at an early stage after main review R1. In the R2' the required DHF and DMR are checked for consistency and completeness, refer to **Chapter 5.5.3**. It should confirm the required inputs are adequate for creating the realization concept. The detailed project management structure and planning are in place.
- after the main review R2, instead of the M200 decision, a M230 decision which covers the content of the M200 decision shall be implemented. The M230 decision shall be made before the main review R3, the FPP and main review R2 are the basis for the M230 decision.

Roles in PLM/产品生命周期管理中的角色

When functional departments/participants are named in this document, it refers to roles, not departments/persons.

Document control/文档控制

For technical documentation controlling, refer to [7].
For customer document controlling, refer to [11].

Patents/专利

Patents must be considered in all process phases of the PEP (applying for patents, observing external patents).

Interfaces between Process phases/两个过程阶段之间的联系接口

When reviewing the process phase results, it is advisable to have a technical expert from the preceding and succeeding or parallel process phase participate, so that errors are avoided, and correct transfer is ensured.

FMEA/失效模式效果分析

In order to systematically inspect and reduce errors, or the possibility of errors, all new or further developments should use the FMEA method in the appropriate scope.

System, construction, or process FMEA's may be used.

产品发布团队（解决方案团队的一部分）负责发布医疗设备产品以及启动开发项目的 M150。产品发布团队（项目团队）负责项目开发中需要符合的质量规定。

如果项目团队需要把评审 R2 推迟到接近评审 R3 的时间，必须符合以下过程：

- 在评审 R1 之后进行额外的评审 R2'。审评 R2'会确认开发历史文档（DHF）和产品制造性文档(DMR)的一致性和完成度，参考 **5.5.3 章节**。评审 R2'需要确认已有足够的信息开始创建产品概念，并建立对应的项目管理和执行计划。
- 在评审 R2 之后用 M230 决定代替 M200 决定。M230 决定必须在审评 R3 之前完成，并覆盖所有 M200 决定的内容。M230 决定基于产品财务计划(FPP)和审评 R2。

当本文档中命名为职能部门或参与者时，它指的是角色，而不是泛指部门或个人。

对于技术文档的控制请参考[7]。对于用户文档的控制，请参考[11]。

在产品设计过程的所有过程阶段中都要考虑到专利问题(申请专利，遵守外部专利法则)。

当评审过程阶段的结果时，建议要有一名技术专家，既参加过前一过程阶段又要参加下一过程阶段或者一直跟着过程阶段，这样可以避免出错，确保正确移交。

为了系统地进行检查以及减少出错，或者说减少出错的可能性，所有新开发项目或进一步开发应在适当范围内采用 FMEA 方法。

系统、**设计**或过程可以使用 FMEA 方法。

Initiation of future development as a result of the change control procedure/在变更管理流程中进行进一步开发

If a change request from the change control procedure result in a further development in accordance with the product engineering process, this is subject too the requirements of the corresponding process section

如果变更管理程序中的变更请求导致需要按照产品设计控制程序进行进一步开发，则该开发工作应当遵循产品设计控制程序中对入口环节所规定的流程要求。

chosen for entry in the PEP.

Development of modular elements/模块化部件开发

The developemtn of modular elements also follows the product development process sections described in this document – from defining the scope of a modular element through to its validation.

To enable a modular element to be re-used in other systems, the documentation relating to the modular element must be complete and definitive. Each modular element has its own DHF doument plan.

模块化部件开发也需要遵循本文档中关于产品设计的流程要求 — 从定义模块化部件的开发内容到部件的设计确认。

为了确保模块化部件在看多个系统产品中的复用，与模块化部件相关的文档必须完整和明确。每一个模块化部件需要建立自己的 DHF 文档计划。

IT security measures (cybersecurity)/IT 安全措施（信息安全）

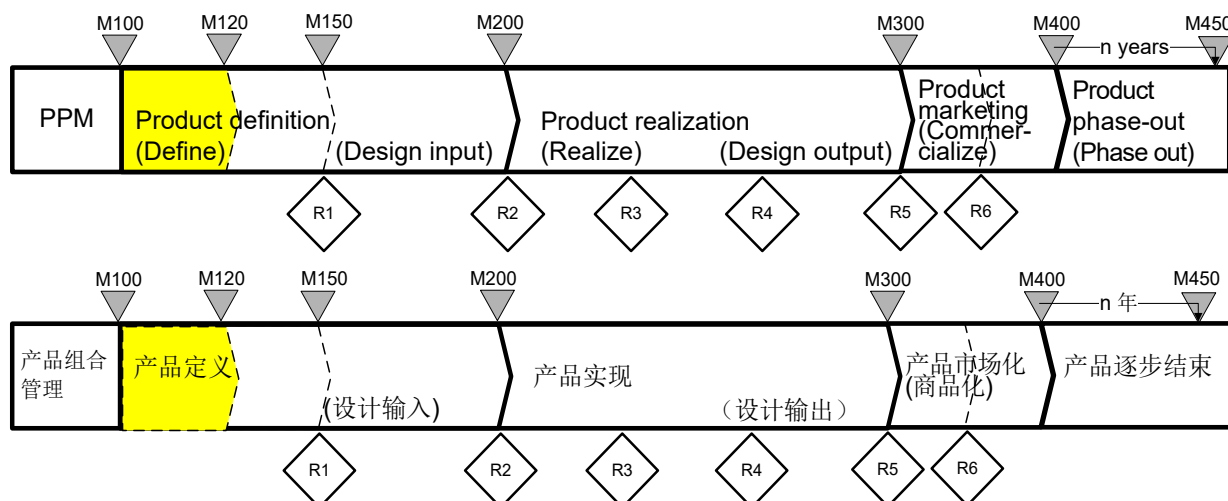
IT security is considered in product development process, detailed process is described in [15].

在产品开发流程中需要考虑 IT 安全措施的实现，具体流程参考 [15]。

5.5 Short description of the process/流程的简单说明

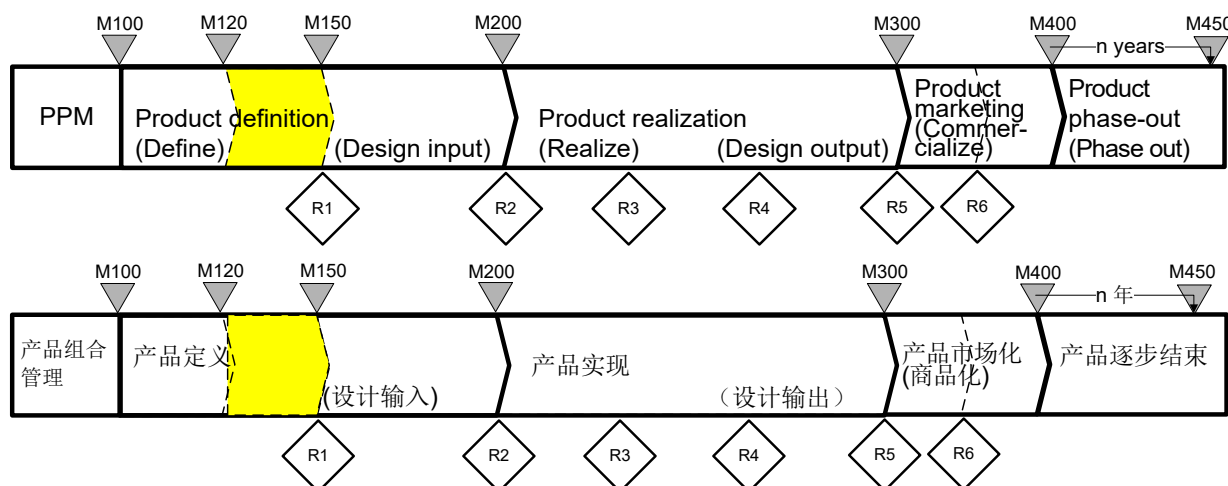
5.5.1 Process section: Product definition – Analyze product-specific market

/过程阶段：产品定义 – 分析产品-特定市场



Start/开始: Decision M100 决定	Basis for decision/决定依据: – Project request M100 – 决定项目是否启动
Purpose of the process section/过程阶段的目的	– Analysis of the product-specific market – Detailed specification of business plan – Start to create PMP (does not need to be a separate document) – 分析产品对应的市场 – 定义具体的商业计划规范 – 开始建立项目管理计划（可以作为其他文档的一部分）

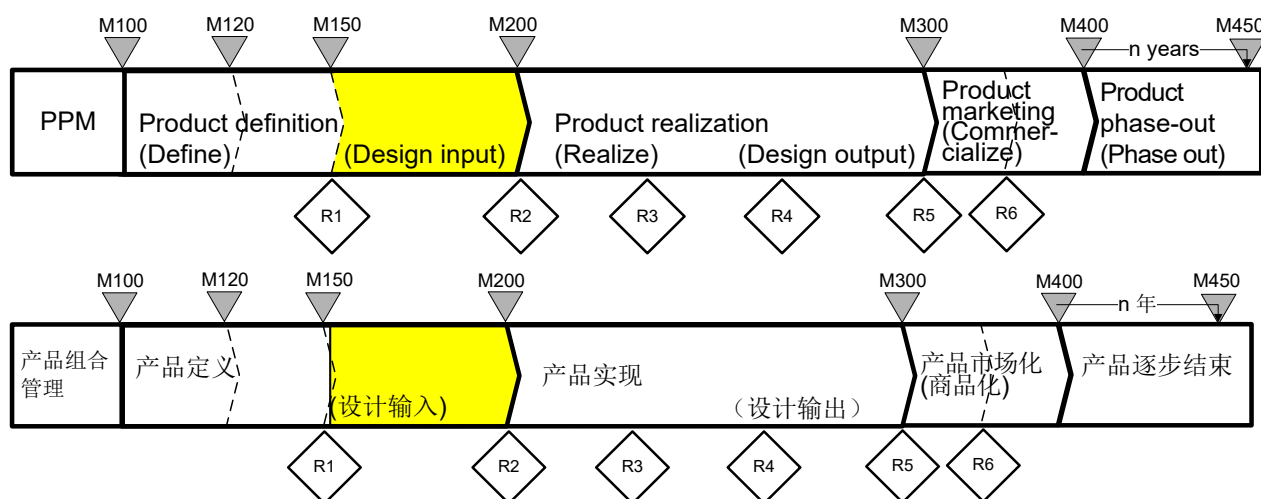
5.5.2 Process section: Product definition – Define marketing and product requirements /过程阶段：产品定义-定义市场和产品需求



Start/开始: Decision M120 决定	Basis for decision/决定依据: – Project request M120 incl. business plan M120 决定项目是否启动
Purpose of the process section 过程阶段的目的	– Determination of customer requirements – Carry out planning for product strategy and product profile – Assessment of clinical performance based on the clinical studies, as part of the input for the PRS, detail see [13] – if necessary, feasibility tests, technology developments – 确定客户需求 – 完成产品战略和概要计划 – 基于临床对临床要求进行评估, 结果作为产品需求规范的一部分, 具体要求见[13] – 在必要时, 进行可行性测试, 技术开发
Activities and Core Deliverables/活动和核心输出	– specify product scope: Product Requirement Specification (PRS) – initial project planning: Project Management Plan (PMP) – initial project planning: Quality Management Plan (QMP) – initial early procurement involvement & early supplier involvement (EPI & ESI) – evaluate product concepts: concept evaluation (serves as input for the functional specification) – A product specific reliability engineering plan is created. – 产品需求规范 – 项目管理计划 – 质量管理计划 – 启动早期采购评估和早期供应商评估 – 评估产品方案 (可以作为功能规范的输入)
Main review R1 (review record) 评审 R1 (评审记录) (refer to chap. 6.1) (参照 6.1 节)	Check of the required DHF and DMR documents for consistency and completeness and confirm that the product requirements are adequately defined. The high-level project management structure and initial project planning are in place. 检查所需的 DHF 和 DMR 文件是否前后一致并完整, 确认产品需求定义, 完成相应的项目管理结构和初始计划 – Product Requirement Specification (PRS) – Creation of the usability engineering file – Define target for environmental and/or health and safety improvement (part of PRS, detail see [14]) – PSS: Product & Solution Security (PSS) project classification – Project Management Plan (PMP) – Quality Management Plan (QMP) – Reliability Engineering Plan – EU MDR: Clinical Evaluation Plan (first version with intended purpose/intended use, clinical claims of the manufacturer, detailed see [13]) – If applicable, DHF and DMR document plan for the " Product definition - Define implementation concept" process section – 产品需求规范 – 建立可用性工程文件 – 定义环境和/或健康安全改善目标 (产品需求规范的一部分, 具体见

	[14]) – PSS: 产品和解决方案（信息）安全项目分类 – 项目管理计划 – 质量管理计划 – 可靠性工程计划 – 临床评估计划（初版需要包括制造商的预期目的/使用和临床声明，具体见[13]） – 根据需要，对“产品定义-定义执行方案”过程计划 DHF 和 DMR 文件 Participants: PM, Quality engineer, product manager, system engineer, any other representatives of functions concerned with the stage of development under review, possibly required specialists 参与者：项目经理，质量工程师，产品经理，系统工程师，必要的研发功能的代表，可能需要相应的专家 An example of the R1 template is shown in appendix 6 for reference. R1 模板见附件 6
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5.5.3 Process section: Product definition – **Define realization concept** /过程阶段：产品定义-定义执行方案



Start/开始: Decision M150 决定	Basis for decision/决定依据: – Project request M150 incl. FPP – QMP and PMP – Product Requirement Specification (PRS) – Review R1 – 项目要求，包括产品财务计划 – 质量和项目管理计划 – 产品需求规范 – 评审 R1
Purpose of the process section/过程阶段的目的	– Creation of functional specification – Creation of architecture on system and on subsystem level – Creation of the risk management plan to document and perform the RM process – Risk assessment and documentation of the risk analysis and risk evaluation – Check the technical feasibility / prototyping (e.g. test prototype)

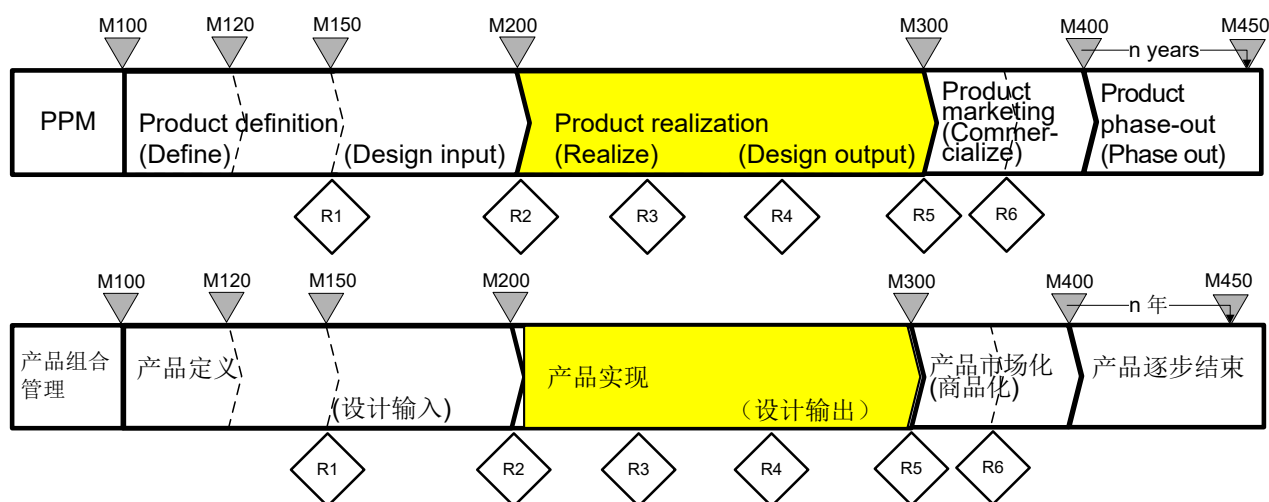
	– Check whether components have already been defined as multiple-use components – Make or buy decision and selection/evaluation of suppliers for purchases – Pre-clarification for developments by third parties (internal/external) – 创建功能规范 – 创建系统和子系统架构 – 创建风险管理计划文档并执行风险管理流程 – 风险评估和风险分析与风险评价文档 – 检查技术可行性/原型研究（例如测试原型） – 检查部件已经被定义为多用途的元件 – 制造或购买的决定，对供应商选择和评价 – 对由第三方开发（内部/外部）的部分进行预澄清
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Activities and Core Deliverables/活动和核心输出	– specify system realization concept: System Functional Specification (SFS) – specify system level architecture: System Architecture Specification (SAS) – specify subsystem architecture: Subsystem Architecture Specification (SSAS) – specify subsystem requirements and realization concept: Subsystem Functional Specification (SSFS) – create product risk analysis: Product Risk Analysis (RA) – plan verification and validation testing: Test Concept – plan Customer Use Test: CUT Plan – plan country approvals: Regulatory Affairs Plan – plan early supplier involvement (EPI planning) For entrusted development, - Sign entrusted development Contract and the quality assurance agreement, where responsibilities at corresponding party, design and development scope and content, as well as relevant technical issues shall be clearly defined. Specifically, the requirements on design and development output documents shall be defined. – Functional Specification of entrusted development should be ready (if applicable) – Define the verification / acceptance scheme of entrusted development in project QMP. Specific verification / acceptance scope shall be determined based on risk analysis and review of supplier verification results, in order to justify whether the design outputs from the entrusted party fulfills the design inputs. – 指定系统实现方案：系统功能规范（SFS） – 指定系统架构：系统架构规范（SAS） – 指定子系统架构：子系统架构规范（SSAS） – 指定子系统的需求和实现方案：子系统功能规范（SSFS） – 创建产品风险分析：产品风险分析（RA） – 计划验证和确认测试：测试方案 – 计划客户使用测试：CUT 计划 – 各国市场准入许可计划：法规事务计划 – 计划早期供应商评估 对于委托研发， – 签订委托研发合同和委托研发质量协议，明确规定各方责任、研发内容及相关的技术事项。其中委托研发质量协议中需定义对于委托设计开发输出文档的要求。 – 委托研发的功能规范文件（如适用） – QMP 中定义委托研发成果验证/验收方案。具体验证/验收范围应基于风险分析和供应商验证结果的评审，以用于判断受托方的设计开发输出是否满足设计开发输入的要求。
Main review R2 (review record) 评审 R2 (评审记录) (refer to chap. 6.2) (参照 6.2 节)	Check of the required DHF and DMR documents for consistency and completeness and confirm that the product realization concept is adequately defined. The detailed project management structure and planning is in place. – System Functional Specification (SFS) – System Architecture Specification (SAS), part of main review R2’ – Subsystem Functional Specification (SSFS) – Subsystem Architecture Specification (SSAS)

	– Risk management plan (RMP), part of main review R2' – Risk assessment (Risk analysis and risk evaluation and acceptance of the total of all residual risks) – Creation of the risk management file – PSS: Creation of the threat and risk analysis (TRA), part of main review R2' – Usability engineering file update – Reliability Concept, part of main review R2' – EU MDR: Clinical Evaluation Plan (with intended purpose/intended use, clinical claims of the manufacturer, pre-clinical evaluation of clinical claims of manufacturer, detailed see [13]), part of main review R2' – EU MDR: Clinical Development Plan (if required, derived from the pre-clinical evaluation of the clinical claims of the manufacturer, detailed see [13]), part of main review R2' – Test Concept (can be linked to the QMP), part of main review R2' – Customer Use Test Plan (can be linked to the QMP) – Regulatory Affairs Plan (can be linked to the QMP) – Documents according to document plans – Project Management Plan (PMP) update, part of main review R2' – Quality Management Plan (QMP) update, part of main review R2' – Partial reviews PR2 – DHF and DMR document plan for the "Product realization" process section, part of main review R2' – Entrusted development Contract and the Quality Assurance Agreement – 检查所需的 DHF 和 DMR 文件的一致性和完整性，确认产品实现方案是否充分定义。详细的项目管理结构和计划。 – 系统功能规范（SFS） – 系统架构规范（SAS），审评 R2'的一部分 – 子系统功能规范（SSFS） – 子系统架构规范（SSAS） – 风险管理计划（RMP），审评 R2'的一部分 – 风险评估（风险分析、风险评价和对剩余风险接受） – 创建风险管理文件 – PSS: 创建威胁和风险分析（TRA），审评 R2'的一部分 – 可用性工程文件更新 – 可靠性工程方案，审评 R2'的一部分 – 临床评估计划（包括制造商的预期目的/使用和临床声明，以及对临床声明的预先临床评估，具体见[13]），审评 R2'的一部分 – 临床开发计划（如果需要，基于对制造商临床声明的预先临床评估内容制定临床开发计划，具体见[13]），审评 R2'的一部分 – 测试方案（可以和 QMP 有链接），审评 R2'的一部分 – 客户使用测试计划（可以和 QMP 有链接） – 法规事务计划（可以和 QMP 有链接） – 根据文件计划所列的文件 – 项目管理计划（PMP）的更新，审评 R2'的一部分 – 质量管理计划（QMP）更新，审评 R2'的一部分 – 子系统评审(PR2)的结果
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	– DHF 和 DMR 文件计划，审评 R2'的一部分 – 委托研发合同和质量协议 Participants: PM, Quality engineer, system engineer, any other representatives of functions concerned with the stage of development under review, possibly required specialists 参与者：项目经理，质量工程师，系统工程师，必要的研发功能的代表，可能需要相应的专家 An example of the R2 template is shown in appendix 7 for reference. R2 模板见附件 7
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5.5.4 Process section: Product realization/过程阶段：产品实现



Start/开始: Decision M200 决定	Basis for decision/决定依据: – Project request M200 incl. FPP 项目要求，包括产品财务计划 – PMP 项目管理计划 – QMP 质量管理计划 – Functional specification 功能规范 – Risk analysis 风险分析 – Risk management plan 风险管理计划 – Review R2 R2 评审
Purpose of the process section/过程阶段的目的	– Development of the product (definition of design, realization, testing) until it is ready for series production in accordance with the functional specification. – Creation of the project-specific product introduction plan – Updating of the risk management file – Continuation of the risk assessment – Risk control – Creation and checking of the documents required for production. – Carry out process and design validation on the production prototype (first batch) – Perform and document the ecological product evaluation. – Perform the tests required for the country-specific approvals on the production prototype (first batch) (e.g. proof of safety, conformity assessment). – Creation of documents required for delivery release in accordance with the DHF and DMR document plan. – 开发产品（定义，设计，实现，测试）以满足功能规范直到它准备好序列生产。

	– 创建的项目相关的产品上市计划 – 风险管理的文件更新 – 持续风险评估 – 风险控制 – 创建和检查生产所需的文件。 – 对生产原型（第一批）进行工艺和设计确认 – 执行并文档化对生态产品的评价。 – 执行国家准入对生产样机（第一批）要求的测试（如安全性证实、一致性评定）。 – 根据 DHF 和 DMR 文件计划建立发货要求的文档
Activities and Core Deliverables/活动和核心输出 Realize and integrate product	– Specify requirements and realization concept on HW and SW components level: HW/SW Functional Specification (HWFS/SWFS) – Create detailed product design: HW/SW design specification – Implement and realize design: HW engineering samples, SW Source code, SW baselines – Create verification test specifications: verification test specifications (according test concept) – Perform SW code reviews: code reviews record – Perform verification testing: verification test records (depending on product structure and test concept: records on HW module/component and SW unit level; includes records for safety testing) – 确定硬件和软件组件级别的需求和实现方案：硬件/软件功能规范（HWFS/SWFS） – 创建详细的产品设计：硬件/软件设计规范 – 实施和实现设计：硬件工程样品、软件源代码、软件基线 – 创建验证测试规范：验证测试规范（根据测试方案） – 执行软件代码审查：代码审查记录 – 执行验证测试：验证测试记录（根据产品结构和测试方案：硬件模块/组件和软件单位水平；包括安全测试）
Main review R3 (review record) 评审 R3 (评审记录) (refer to chap. 6.3) (参照 6.3 节)	Check of the required DHF and DMR documents for consistency and completeness and confirm completion of design and development – HW/SW Functional Specification (HWFS/SWFS) – HW/SW design specification (HWDS/SWDS) – Specifications for ▪ Single function tests (HW module and SW unit) ▪ Subassembly integration tests / SW component integration tests (HW component and SW item) ▪ Subsystem integration tests / SW subsystem integration tests ▪ System integration tests – Reports on the specified tests and reviews, e.g. SSIT Records before System R3 – Input for partial / main review of the level above: Release note with information about implemented requirements, not implemented requirements, performed tests (including summary of test result), not performed tests, known bugs/restrictions (e. g. main review needs the release notes of all subsystems) – Risk management plan (RMP) update – Risk assessment update and risk control (Risk analysis and risk evaluation and acceptance of the total of all residual risks)

	– Risk management file update – PSS: Threat and risk analysis (TRA) update – Usability engineering file update – Define IVK list (system level) – Define spare part list – Documents according to document plans – Partial reviews PR3 检查所需的 DHF 和 DMR 文件的一致性和完整性。确认设计和开发完成 – 硬件/软件功能规范(HWFS/SWFS) – 硬件/软件设计规范(HWDS/SWDS) – 以下规范： • 单功能测试（硬件模块和软件模块） • 组件集成测试/软件组件集成测试（硬件组件和软件组件） • 子系统集成测试/软件子系统集成测试 • 系统集成测试 – 指定的测试和评估报告，如在系统 R3 前完成的子系统集成测试记录 – 对系统/子系统评审的输入：发布说明需要包括：实现的需求，没有实现的需求，完成的测试（包括测试结果的总结），没有完成的测试，已知错误/限制（系统评审需要所有子系统的发布说明） – 风险管理计划（RMP）更新 – 风险评估的更新和风险控制（风险分析、风险评价和对剩余风险接受） – 风险管理文件更新 – PSS: 威胁和风险分析（TRA）更新 – 可用性工程文件更新 – 定义 IVK 列表（系统级） – 定义备件列表 – 根据文件计划的文件 – 子系统评审 PR3 Participants: PM, Quality engineer, system engineer, any other representatives of functions concerned with the stage of development under review, possibly required specialists 参与者：项目经理，质量工程师，系统工程师，必要的研发功能的代表，可能需要相应的专家 An example of the R3 template is shown in appendix 8 for reference. R3 模板见附件 8
Activities and Core Deliverables/活动和核心输出 Verify product	– Perform final verification: verification test records (according test concept: records on subsystem/system level; includes test records for safety testing) and Verification Test Report(s) (summary reports) – Create validation test specifications: validation test specifications – Confirm completion of HW verification activities: engineering release notes and record of partial review R4 – Confirm completion of SW verification activities: SW baseline for partial review R4 and record of partial review R4 For entrusted development by supplier: – Review and archive the outputs from supplier in SSME XP, according to required

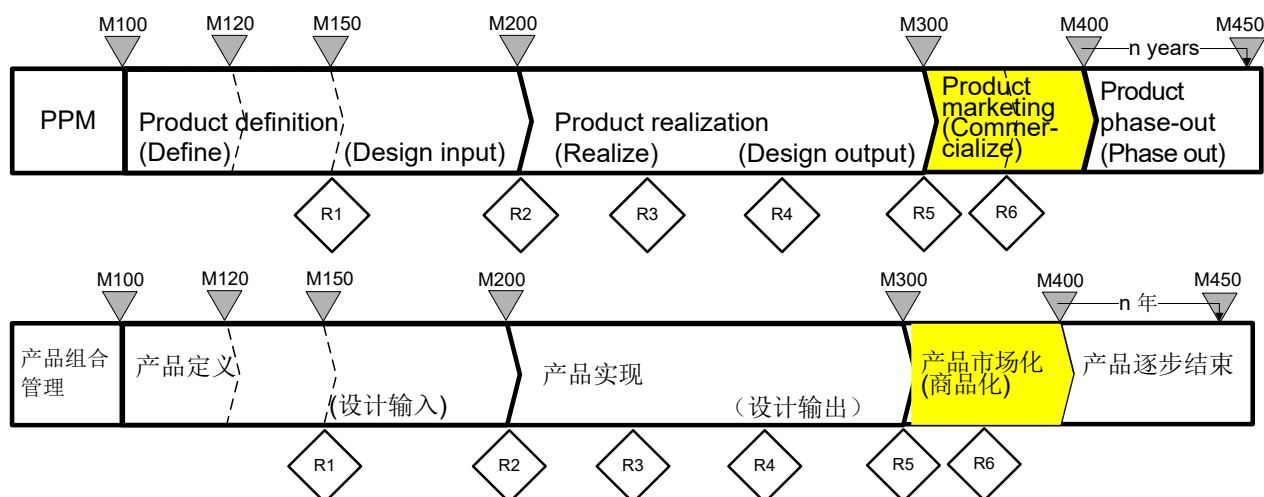
	development output list as defined in Development Agreement – Validate/Verify the entrusted system/subsystem/component – 执行最后验证：验证测试记录（根据测试方案：子系统/系统的记录；包括安全测试记录）和验证测试报告（总结报告） – 创建确认测试规范：确认试验规范 – 确认硬件验证活动完成：研发发布说明和子系统 R4 评测记录 – 确认软件验证活动完成：软件子系统 R4 基线定义和评测记录 对于委托研发： – 根据委托研发的设计输出文件清单对设计输出进行评审和存档 – 完成委托研发系统 / 子系统 / 部件的确认/验证
Main review R4 (review record) 评审 R4 (评审记录) (refer to chap. 6.3.4) (参照 6.3.4 节)	Check of the required DHF and DMR documents for consistency and completeness and confirm completion of product verification activities (includes the evaluation of the system integration tests) – Reports on the specified tests and reviews – Verification Test Report(s) (summary reports) – Validation test specifications – Product introduction plan – Risk assessment update and risk control update (Risk analysis and risk evaluation and acceptance of the total of all residual risks) – Risk management file update – PSS: Threat and risk analysis (TRA) update – Usability engineering file update – Special parts (Identification of S and R parts that are subject to mandatory monitoring, Definition of IVK parts) – After completed partial reviews for R4, transfer reports for the units to be transferred – Documents according to document plans – Partial reviews PR4 – Output from supplier (if applicable) Check whether licenses are available for the third-party software used 检查所需的 DHF 和 DMR 文件的一致性和完整性，确认完成产品验证活动（包括系统集成测试的评估） – 指定的测试和评估报告 – 验证测试报告（总结报告） – 验证试验规范 – 产品上市计划 – 更新风险评估和风险控制（风险分析、风险评价和对剩余风险接受） – 风险管理文件更新 – PSS: 威胁和风险分析（TRA）更新 – 可用性工程文件更新 – 特殊部件（对 S 和 R 部件识别和强制性监测，定义 IVK 部件） – 完成子系统评审 PR4，移交部件的报告 – 文件计划中的文件

	– 子系统评审 PR4 – 供应商的设计输出（如适用） 检查所使用的第三方软件的许可证是否可用 <u>Participants:</u> PM, Quality engineer, system engineer, any other representatives of functions concerned with the stage of development under review, possibly required specialists 参与者：项目经理，质量工程师，系统工程师，必要的研发功能的代表，可能需要相应的专家 An example of the R4 template is shown in appendix 9 for reference. R4 模板见附件 9
Activities and Core Deliverables/活动和核心输出 Prepare supply chain Accept product	– Produce first batch: First batch production incl. process validation (as well as the production review reports) – Perform validation tests: Reports for validation tests, test reports for validation (summary), risk management report, test report for usability (summary) – Prepare general availability and unlimited shipment: User documentation, SCM documentation, CS documentation, clinical evaluation – Provide country specific approvals: Country specific approvals – Perform customer use test: Report for customer use test – MDD: Create technical file according to MDD and perform a review: technical file – EU MDR: Create technical file according to MDR and perform a review: technical file – EU MDR: Complete clinical evaluation for the release of the medical device and define post-market surveillance activities – Check CE conformity assessment: CE declaration of Conformity – 首批产品生产：首批生产，包括工艺/流程验证（以及生产审查报告） – 执行确认测试：确认测试报告，确认试验报告（摘要）、风险管理报告、可用性测试报告（总结） – 为发货做准备：用户文档，生产文档，客户服务文档，临床评估 提供特定国家的批准：特定国家的准入许可证 – 执行客户使用测试：客户使用测试报告 – 根据 MDD 要求创建技术文件并进行审查：技术文件 – 根据 MDR 要求创建技术文件并进行审查：技术文件 – 根据 MDR 要求完成医疗设备上市的临床评估并且定义后市场监管活动 – 检查 CE 合格评定：CE 符合性声明
Main review R5 (review record) 评审 R5 (评审记录)	Check of the required DHF and DMR documents for consistency and completeness and confirm completion of design validation and that the product is ready for the release to general availability – Reports for the specified tests and reviews (e.g. system test, acceptance test, usability) – Test reports for validation (summary) – Risk control (documentation of evaluation of residual risk and acceptance of the overall residual risk) – Risk management report – Released risk management file – PSS: Product & Solution Security (PSS) Residual Cybersecurity Risk Report – Released usability engineering file

	– Reliability Report – Documentation of the ecological product evaluation – User documentation: e.g. operator manual incl. data sheet – SCM documentation: supplemental production documents (e.g. work plans, test instructions, incl. documentation for production and test equipment) – First batch production incl. process validation (as well as the production review reports) – CS documentation: Technical documentation, e.g. service documentation, maintenance instructions and template of the installation report – Assessment of a restriction list with the requirements not implemented from the product requirement specification and/or functional specification, system flowchart and price book, including the shortcomings of the system. – EU MDR: Clinical Evaluation Report (depending on the applicable regulation, detailed see [13]) – EU MDR: Post-market Surveillance Plan – EU MDR: Post-market Clinical Follow-up Plan (can be part of the clinical evaluation plan, detailed see [13]) – Report of the customer use test – Assessment of the customer use test – Documents according to document plans (DHF/DMR) – Check whether the country-specific approvals required are available – MDD: Proof of Essential Requirements – EU MDR: Proof of General Safety and Performance Requirements – MD: Conformity with Essential Health and Safety Requirements (depending on the applicable regulation) – MDD: Technical file according to MDD – EU MDR: Technical file according to MDR – If applicable updated project management plan (PMP) – Reviewed and if applicable updated quality management plan (QMP) with monitoring plan and the quality key performance indicator – Partial reviews PR5 – 检查所需的DHF和DMR文件的一致性和完整性，确认完成设计验证和产品准备发布 – 报告指定的测试和评估（例如，系统测试，验收测试，可用性） – 确认测试报告（总结） – 风险控制（对剩余风险评估和接受并文档化） – 风险管理报告 – 发布风险管理文件 – PSS: 产品和解决方案（信息）安全剩余风险报告 – 发布可用性工程文件 – 可靠性工程报告 – 对产品生态环境的评估文档 – 用户文档：如操作手册包括数据表 – 生产文档：要求的生产文件（如工作计划、测试指导书，包括生产和测试设备文档） – 首批生产检验，包括工艺/流程（以及生产审查报告） – 客户服务文件：技术文档，例如服务文档，维护说明书和安装报告模板 – 对限制列表进行评估，包括没有实现的产品需求和功能规格，系统流程图和价格书，以及系统的缺陷。 – EU MDR: 临床评估报告（取决于适用的法规，具体见[13]） – EU MDR: 后市场监督计划
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	- EU MDR: 后市场临床应用追踪计划（这部内容也可以包括在临床评估计划中，具体见[13]） - 客户使用测试报告 - 客户使用测试结果的评估 - 根据文件计划（DHF / DMR）要求的文件 - 检查特定国家审批结果 - MDD: 证明产品满足基本需求（ER） - EU MDR: 证明产品满足通用安全和性能需求（GSPR） - 根据MDD创建的技术文件 - 根据MDR创建的技术文件 - 如果适用，更新项目管理计划（PMP） - 如果适用，更新质量管理计划并审查（QMP），包括定义需要监控的质量指标 - 子系统评审PR5 Participants: PM, Quality engineer, system engineer, any other representatives of functions concerned with the stage of development under review, possibly required specialists 参与者：项目经理，质量工程师，系统工程师，必要的研发功能的代表，可能需要相应的专家 An example of the R5 template is shown in appendix 10 for reference. R5 模板见附件 10
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5.5.5 Process section: Product marketing /过程阶段：产品市场化



Start/开始: Decision M300 决定	Basis for decision/决定依据: - Project request M300 incl. FPP 项目要求，包括产品财务计划 - Review R5 R5 评审
Purpose of the process section/过程阶段的目的	- Placing the released products on the market - Ensuring support, maintenance and service - Extension/confirmation of country-specific approvals - Risk management; systematic evaluation of information about delivered medical devices during product marketing - 发布产品到市场上 - 保障支持、维护和服务 - 扩展/确认各国市场准入许可

	– 风险管理；在产品发布过程中对已经发货的产品进行系统性的评估
Activities and Core Deliverables/活动和核心输出	– Product monitoring – Takes appropriate measures (corrective and preventive actions) – 产品监控 – 采取适当的措施(纠正和预防措施)
Main review R6 (review record) 评审 R6 (评审记录)	Assessment of the quality key performance indicators with respect to quality objectives in the QMP (if applicable) including specifications from the product introduction plan and reliability plan. 根据 QMP (如果适用)包括产品上市计划和可靠性工程计划定义的规范，对关键质量性能指标进行评估。 <u>Participants:</u> PM, Quality engineer, system engineer, any other representatives of functions concerned with the stage of development under review, possibly required specialists. 参与者：项目经理，质量工程师，系统工程师，必要的研发功能的代表，可能需要相应的专家

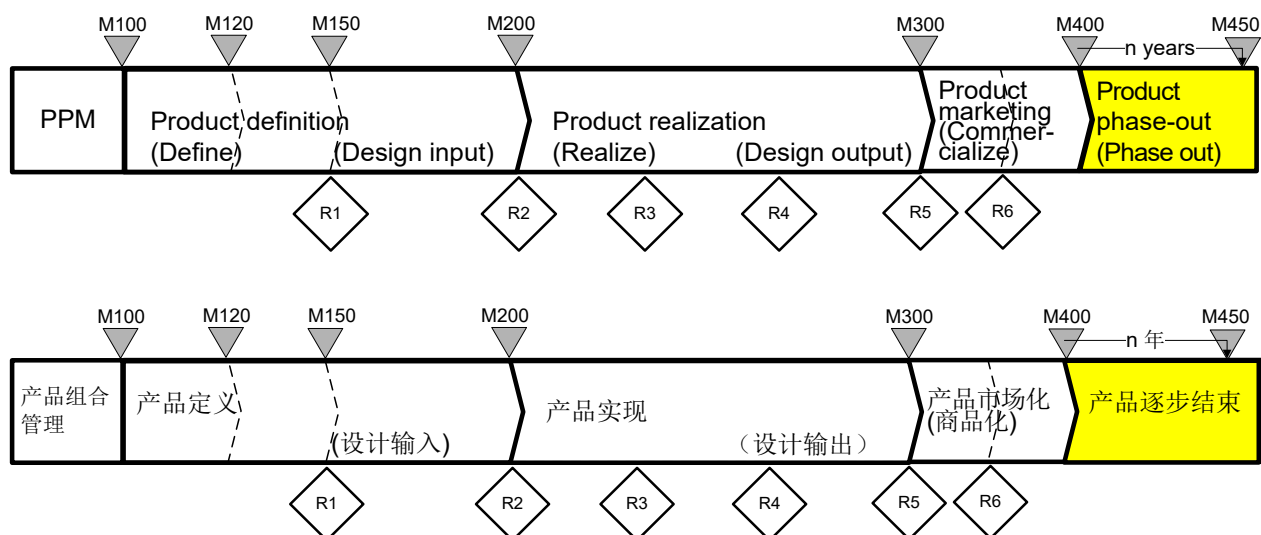
If the product launch plan or PMP does not specify a date for main review R6, the review should be performed no later than 12 months after M300.

If main review R6 determines that the requirements have not been achieved, the project manager initiates escalation to the service manager, the R&D manager, the SSME XP head of Quality Management and the head of the **SSME XP in BIT meeting**.

如果在产品上市计划或者项目计划中没有指定 R6 主评审的日期，默认 R6 主评审日期不晚于 M300 后 12 个月。

如果在 R6 时发现相应的要求没有达到，项目经理应该在 **BIT 会议上**向服务经理，研发经理，SSME XP 的质量总监和总经理做出汇报。

5.5.6 Process section: Product phase-out /过程阶段：产品逐步结束



Start/开始: Decision M400 (end of delivery)/M400 决定 (结束发货)	Basis for decision: /决定依据 – Project request M400/项目要求 M400
Purpose of the process section/过程阶段的目的	– Ensuring support, maintenance and service until end of spare parts delivery – Risk management; systematic evaluation of information about delivered medical devices during product phase-out – 在停止供应备件之前，保证产品的支持、维修和服务 – 风险管理; 在产品结束过程中对已经发货的产品进行系统性的评估

5.6 Structure levels of systems/系统的结构层次

Generally, systems are comprised of the system levels displayed in Fig. 5. If necessary, structure levels may be omitted or added.

一般，系统是由图 5 中所列出的各个系统层次组成的，可以根据需要省略或添加结构层次。

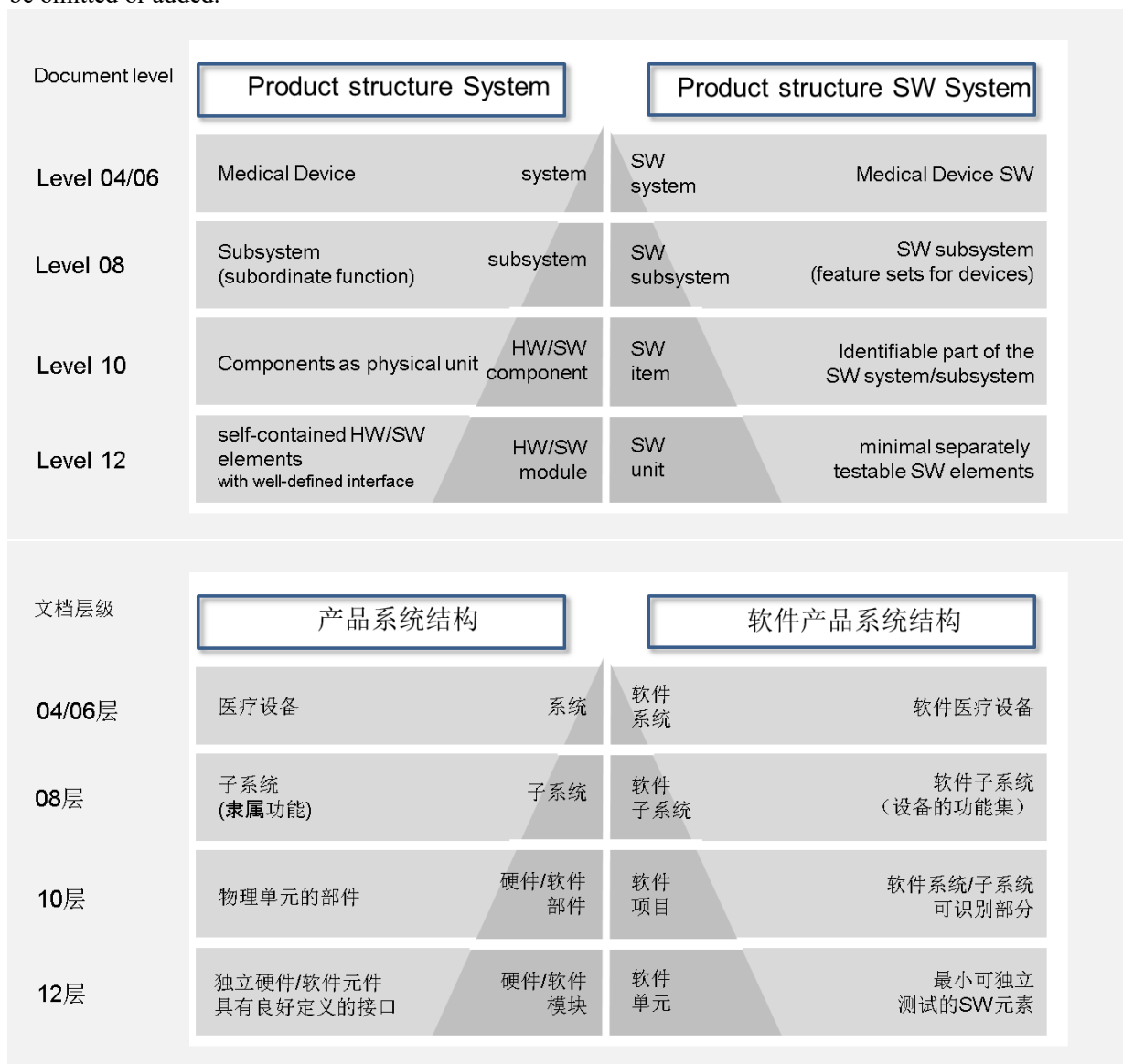


Fig. 5/图 5: Structure levels of systems/系统的结构层次

Systems, sub-systems and components similarly follow the V-model.

Examples for applying the V-model to sub-systems are contained in appendix 1.

Depending on the order, the vendors generally conclude their projects with the given main review.

Acceptance can result through the participation of the person awarding the order in the successful test.

系统、子系统和部件同样要遵循 V 字型模式。

附录 1 含有子系统运用 V 字型模式的实例。

供货商通常根据订单通过主评审才算完成项目。

根据订单的状态，并基于通过的主评审结果，供应商对其负责的项目作相应决定。

6 In-depth process description / 深化过程说明

6.1 Process section: Product portfolio management / 过程阶段：产品组合管理

Product portfolio management supports the prioritization of investment in products and service that are consistent with the needs of our customers, our strategy and our business objectives.

产品组合管理确保对产品和服务的投资优先级与客户需要、战略目标和业务目标一致。

6.2 Process section: Product definition – Analyze product-specific market / 过程阶段：产品定义 – 分析产品对应市场

If a product idea is followed up, this phase is started with an M100 decision of the management.

管理层通过 M100 的决定启动对一个产品想法的进一步分析。

The product-specific market is analyzed, and the business plan is specified in detail.

内容包括分析产品对应市场，准备具体商业计划。

Between M100 and M120 the product manager compiles the required results together with an interdisciplinary team. The PSG is designated as part of the M120 decision.

在 M100 到 M120 之间，产品经理与一个跨职能团队一起整合分析内容。M120 的决定包括指定 PSG 团队。

6.3 Process section: Product definition – Define marketing and product requirement / 过程阶段：产品定义 – 定义市场和产品需求

The product definition process should preferably employ integrated product planning (IPP includes a methodically structured procedure for defining requirement specifications and functional specifications), which allows you to achieve market- and customer-oriented implementation of strategic business objectives, shorter planning times, higher planning quality and a more efficient method for creating requirement specifications and functional specifications.

产品定义流程应该使用整合产品计划（整合产品计划 IPP 是一套结构化定义需求规范和功能规范的过程），通过整合产品计划可以实现面向市场和客户的战略商业目标，更短的计划时间，更高的计划质量以及更高效地创建需求和功能规范。

The IPP consists of three sequential subphases: **product strategy**, **product profile** (to the product requirement specification) and **product concept planning** (to the functional specification) with the following objectives, analyses and results:

IPP 有三个连续的过程构成：**产品策略**，**产品概要**（需求规范），和**产品方案计划**（功能规范），其主要的目标，分析和结果包括：

- To define **product strategy**, in which the main success factors of the new *product*, the target markets, customer groups, its strengths and weaknesses and any restrictions are analyzed and defined.
- To define **product profile** by determining, defining, and prioritizing:
 - Market and customer requirements (from the customer in the customer's language) for products and "Customer Service & Support" services
 - Historical data of predecessor products (e.g. reportable events, complaints, results of fault investigations)

- 制定**产品策略**，需要分析和定义新产品的成功因素，目标市场，客户群体，优势和劣势，以及局限性。
- 制定**产品概要**，应该决定，定义和取舍：
 - 产品的市场和客户需求（用客户语言描述），以及客户服务与支持的需求。
 - 分析前期产品的历史数据(如医疗设备事件汇报、抱怨、出错调查结果)。

➤ To define the **product concept** through

Prioritizing and defining the function, component and service structure, taking into consideration multiple use, modularity and standardization, the systematic implementation of which into technical characteristics (product functions / target values for development and production) should result from QFD analyses, for example,

- of the material, process and product technologies, and the technologies of competitor products
- of the development and production resources, and the budget
- Consideration of design.

➤ 制定**产品方案**，需要

取舍和定义功能、部件和服务结构，考虑复用、模块化和标准化，系统化实现技术特性（产品功能/开发和生产指标）并进行 QFD 分析，比如：

- 分析物料，工艺流程和技术方案，以及竞品采用的技术方案。
- 确认开发和生产的资源，以及预算
- 考虑设计

6.3.1 Product requirement specification / 产品需求规范

PRS and FS (in chap. 6.4) are a part of design input documents. They describe the required product **features** in a complete and detailed manner, so that they can later be used as the basis for product development, as well as for design validation.

The PRS contains product **features** from the user's perspective (intended use) and is typically generated at system level, under the **overall control of product management**.

Product feature must be provided in the form of concrete values or value ranges, including tolerances, if applicable, which must be refined and narrowed down in subsequent documents.

The PRS also includes information regarding:

- The target markets and the resulting statutory and normative requirements for country-specific approvals
- The service concept (e.g. serviceability requirements, target for key performance indicators such as mean time between failures)
- Requirements from the usability process
- Requirement for product reliability as well as the expected service life, for example
- Requirement for data security, if applicable
- Requirement for a network, if applicable

产品需求规范和功能规范(6.4 节所述)是设计输入文件的一部分。它们对所要求的产品特性做完整详细的说明，因此它们可以作为以后产品开发和设计确认的依据。

产品需求规范包括以用户观点提出的产品特性 (预期用途)，通常由**产品管理部门**，根据系统级别的需求编写的。

必须提供产品特性，**包括具体值或值的范围的形式**，必要时包括公差，这些信息必须在随后的文件中进一步确定下来。

产品需求规范也包括：

- 目标市场和与特定国家上市相关的法律法规要求
- 服务方案 (包括可服务性需求，关键服务指标的目标，如平均故障间隔时间 MTBF)
- 可用性流程规定的要求
- 产品可靠性需求以及预期的服务寿命
- 如果适用，数据安全的需求
- 如果适用，网络需求

6.3.2 Design planning / 计划设计阶段

Design planning takes place via the quality management plan (QMP) and project management plan (PMP). Depending on the project, a configuration management plan and a test concept are produced, possibly together with other documents.

计划设计阶段通过质量管理计划(QMP)和项目管理计划(PMP)开展。根据项目的具体内容，可以创建可配置的管理计划，测试方案和其他文档也可以包含在管理计划中。

Quality management plan (QMP)

The QMP is a summary of project specific quality-relevant directives. It contains the detail procedure for the project, the following items shall be described into the project QMP:

- Project objectives and entry in development projects (in case of changes);
- Determination of the development phases and milestones, and the planning of these;
- The project organization including the functional departments and responsibilities and the definition of interfaces to other organizational units;
- Affected projects (if present) and interfaces between these projects;
- Composition of the PSG (roles);
- Product structure plan (a predecessor of system flowchart), comprises the scope of delivery/service of the product, its subsystems and assembled components organized hierarchically including options and accessories;
- Document planning including document structure, review planning and traceability concept;
- Type and number of reviews and tests, including the participants (roles, reference to test concept, if necessary);
- Classification according to MDD and/or MDR (depending on the applicable regulation);
- Definition of the type of clinical evaluation;
- The test concept;
- The number of production prototypes required for process validation (including the reason for the selection);
- Procedure for archiving source code (if applicable);
- The criteria for terminating and repeating process phases;
- Procedure for the customer user test (if the project doesn't require a CUT, a reason must be provided. The change of the procedure must be accepted in M200 decision);
- Quality management of products and services provided by third parties (including activities for tracking and controlling outsourced development services);
- Reason for deviating from the procedure of the present regulations (e.g. tailoring);

质量管理计划 (QMP)

质量管理计划是质量相关指令的关于项目的总结。它包含项目所需的详细程序，以下条目需在质量管理计划中描述：

- 项目目标和开始阶段（适用变更管理）；
- 决定开发阶段的里程碑以及相应计划；
- 项目组织，包括职能部门，职责以及明确定义跟其他组织机构的接口关系；
- 受影响的其他项目（如果有）以及与这些项目的接口关系；
- PSG 的组成（角色）；
- 产品结构计划（系统流程图的前身），包括产品以及按照测试排列的各个子系统，部件和模块及软件单元的供应和服务的范畴；
- 文档计划包括文档结构，审评计划和需求追溯方案；
- 审评和测试的类型和数量，包括参与者（角色，如果必要可以参考测试方案）；
- 根据 MDD 和或 MDR 确定项目的类别（取决于适用的法规）；
- 定义临床评估的类型；
- 测试方案；
- 过程确认所需的生产样机的数量（包括选用原因）；
- 存储源代码的程序（如果适用）
- 结束和重复过程阶段的准则；
- 客户应用测试的程序（如果项目不需要进行客户应用测试，必须提供理由，并且包括在 M200 决定中）；
- 由第三方提供的产品或服务的质量管理（包括对外包活动的进度追踪和控制）；
- 说明修改程序与当前程序规定不同的理由（如量体裁衣）；

- Project-specific procedure for concurrent engineering, if necessary;
- Procedure for ensuring the acquisition of country-specific approvals;
- Release planning, version planning, and upgrade strategy, if applicable.

A QMP must be updated for projects up to milestone M300. From milestone M300 onwards, the QMP may still be maintained to describe project-specific regulations. If the QMP is not further updated after milestone M300, it must be checked in the main review R5 whether all project-specific regulations and definitions have been completed.

- 如果必要，项目特定的同步开发程序；
- 确保国家上市相关需求的程序；
- 如果适用，发布计划，版本计划和升级策略。

在 M300 之前，项目 QMP 必须持续更新。在 M300 之后，QMP 仍然可以持续更新，用来描述项目相关的规定。如果决定 QMP 在 M300 后不再更新，那么必须在主审评 R5 时检查所有项目相关的规定和定义已经全部完成。

Project management plan (PMP)

The PMP documents part of the design planning. After M120 it is based on the QMP as well as on the DHF/DMR document plans. It describes the project in detail and includes in particular:

- The time schedule with milestones and main activities (with the required level of detail);
- The work packages in the critical path.

As a separate document, the PMP is an aid for project management. It can be linked to the QMP, depending on the project.

The PMP (with planning of at least the next phase) must be archived at least at the time of each main review.

项目管理计划 (PMP)

项目管理计划是计划设计的一部分。在 M120 之后，PMP 基于 QMP 和 DHF/DMR 文档计划制定。PMP 描述了项目的详细信息，包括：

- 包括里程碑和主要项目活动的时间表（有必要的细节程度）；
- 在关键路径上的工作任务。

PMP 是项目管理的辅助，PMP 可以是独立的文档，也可以合并到 QMP 中，取决于项目的决定。

PMP（至少需要包括下个阶段的计划）最晚需要在每一个主审评之前存档。

6.3.3 DHF / DMR / 设计追溯性文档/产品制造性文档

As part of the project, the DHF and DMR are created and maintained for every product / product group. Once the documents have been entered, the document plans become tables of contents for DHF and DMR. An example of the DHF and DMR index list is shown in appendix 5 for reference.

Document plans:

- Refer to the document plans of sub-systems, components, modules and parts, including purchases, which together form the product,
- Refer to all quality-related documents;
- Identify all documents required for CE marking via a separated ID
- Are project-specific;
- Are similar to a checklist, in that they serve as the basis for the relevant main review,
- Are maintained throughout the whole life-cycle of the product.

Table of contents:

- Refer to the actual location of documents.

在项目过程中，各产品/产品组应制订和维护设计追溯性文档(DHF)和产品制造性文档(DMR)。一旦有文件输入,文件计划即成为 DHF 和 DMR 的目录。附录 5 列出了 DMR/DHF 文件列表模板供参考。

文件计划:

- 针对结合在一起形成产品的所有子系统、部件、组件和部件，包括采购的文档计划；
- 针对所有与质量有关的文件；
- 通过独自的 ID 标识 CE 标志的相关文件；
- 与项目相关；
- 类似于检查列表，可以作为各相应的主评审的依据；
- 在产品的整个生命周期都需要维护。

目录:

- 文档的实际存储位置

The relevant document manager is responsible for drawing up the document plans.

The **DHF** incorporates all documents or types of document which support the development process, for example:

- Design specifications;
- Project management documents (i.e. project applications, QMP/PMP);
- Complete test documentation.

The **DMR** lists:

General documents that are not relevant to the product, yet which contain additional activities that must be observed, such as:

- Q regulations
- Documentation for data management systems that affect quality.

Product-specified (or product family specific) documents required for:

- manufacturing, including meeting environmental regulations;
- distribution (sales, delivery);
- operation and;
- Maintenance.

And

- All lower-level DMRs of the subsystems.

The DHF and DMR document plans can be amalgamated.

The responsibility for the DHF/DMR documents lies with the project manager.

由相应的文档经理负责制定文档计划。

DHF 中列出所有的文件或文件类型，支持开发流程，如：

- 设计规范；
- 项目管理文件(如项目请求，质量管理/项目管理计划)；
- 全套测试文档。

DMR 中列出：

产品通用要求文件，如：

- 质量规定；
- 会影响产品质量的文档数据管理系统。

产品指定的（或产品家族指定的）文件，用于：

- 生产，包括达到环境规范；
- 发售(销售，发运)；
- 运作；
- 保养。

以及

- 更低层级的子系统 DMRs

DHF 和 DMR 文档计划可合并操作。

DHF/DMR 文件由项目经理负责。

6.4 Process section: Product definition – Define realization concept /过程阶段：产品定义 - 定义实现方案

6.4.1 Functional Specifications / 功能规范

The **functional specification** (including the SW requirements specification, where applicable) converts the requirements in the product requirement specification into technical product concepts, and is typically generated under the overall control of the responsible development department. The main components of the functional specification generally are:

- The system architecture, including the SW architecture;
- The functional structure;
- The description of internal and external interfaces, including user interface and service interface;

功能规范（如果适用，也包括软件需求规范）将产品需求规范转化成产品技术方案，通常由开发部门负责创建。

一般地，功能规范的主要组成部分是：

- 系统结构，包括软件结构；
- 功能结构
- 内外部联系接口说明包括用户接口及服务接口；

- The types of measures form the risk analysis for risk control;
- The fulfillment of the legal and normative requirements to get country-specific approvals, while also considering the inspection office which is responsible for later testing;
- Implementation of environmentally friendly product design, including the disposal concept;
- Establishment of measures to realize the service concept

The creation of the functional specification should be preceded by the selection and evaluation of technical product concepts with respect to:

- Time to market
- Target cost
- Target quality

Initially, functional specifications are DHF documents. Following successful design transfer, versions of the functional specifications are DMR documents and thus subject to the change control procedure, refer to [1].

6.4.2 Risk Management / 风险管理

Risk management covers the technical risk analysis and risk assessment. The requirements in terms of the generation and handling of risk management are described in [8].

The risk management plan must be drawn up during the product definition process phase in order to incorporate any necessary measures into the specification documents. It must be continued in subsequent process stages.

Before the product is released, generally by the time of the system integration test, the effectiveness of the measures must be proven and documented in the project documentation. For design measures encompassing all design and test stages, proof of traceability must be provided.

If, in the scope of further product development or product support, the causes of risks are new or changed, the Risk Management file must be supplemented or corrected.

6.4.3 Product introduction plan / 产品上市计划

In order to ensure a systematic, result and target-oriented product introduction, a project-specific product introduction plan must be created. At the M200 decision, the project management requests the necessary resources as well as the budget, and names the members of the product launch team.

• 风险管理中控制风险的措施;

- 为了通过相关国家市场准入的审批程序,其规定的法律和标准需求应在文件中反映,也应将以后负责测试的检验机构考虑进去;
- 进行有利于环保的产品设计包括产品处理方案;
- 为了实现服务方案所建立的措施。

选择和评估产品技术方案,创建功能规范是应该考虑:

- 上市时间
- 成本目标
- 质量目标

功能规范创建之初是 DHF 文件。在完成设计转移后,相应版本的功能规范文件是 DMR 文件,因此需要遵循变更控制程序,参考[1]。

风险管理包括技术风险分析和风险评估。生成和处理风险管理的需求在参考文件[8]中描述。

风险管理计划必须在产品定义流程阶段生成,以便在规范文档中合并必须的方法。而且必须在贯穿于后续的各个流程阶段。

产品发布之前,一般在系统集成测试之前,必须证实所采取的措施是否适用并以文件形式写在项目文件中。对于包含所有设计和测试阶段的设计措施,必须提供可追溯性的证明。

如果在产品进一步开发和产品支持范围内,风险起因是新的或作了修改,必须对风险管理文件进行补充或纠正。

项目需要创建对应的系统化的和结果目标导向的产品上市计划,以确保产品上市成功。在 M200 决定时,项目管理团队提出必要的资源和预算需求,以及产品发布团队的人员名单。

In the phase until main review R4, the product introduction plan must be created and released by the project team. Implementation can be started following the R4 main review and may extend beyond the M300 decision.

6.4.4 Purchases / 采购

If, in the framework of project processing, units or services are purchased internally or externally, please refer to [2].

An early involvement of procurement (EPI) and the supplier (ESI) in the concept phase of development yields considerable advantages for technical conception, cost and time to market.

If, the EPI is necessary, the EPI planning shall be ready for M150.

在主审评 R4 时，项目团队必须创建和发布产品上市计划。计划的实施从 R4 后开始，并可能延长至 M300 之后。

如果在项目处理计划中，单元部件或服务由内部采购或外部采购，请参见[2]。

早期采购和供应商评估可以带来包括技术方案，成本以及上市周期方面的优势。

如果需要启动早期采购评估，那么早期采购评估计划需要在 M150 完成。

6.4.5 Product-related environmental protection / 产品相关的环境保护

The requirements with regard to product-related environmental protection are aimed at the efficient implementation of product-related rulings and concern in particular:

- The identification, restriction and documentation of product ingredients
- Specific requirements for the procurement process
- Environmentally related requirements for product features and
- Requirements for supplementing accompanying documents regarding environmentally related information.

产品需要实现有效的环境保护规定，具体需求包括：

- 识别、限制和记录产品的组成
- 相应的采购流程
- 相应的产品功能
- 相应的环境影响信息补充文件

6.4.6 Procurement and use of third-party software / 采购和使用第三方软件

If third-party software is used, related procedure should be followed, refer to [20].

如果使用了第三方软件，需要遵循相关程序，参考[20]。

6.4.7 Product Reliability / 产品可靠性

The Product Reliability Management process shall minimize negative effect for users or customers due to limited reliability, availability or maintainability of a product.

The reliability related product requirements are handled through the development-V.

For additional information see [17]

产品可靠性管理流程应尽量减少由于产品的可靠性、可用性或可维护性有限而对用户或客户的负面影响。

相关产品的可靠性要求通过开发-V 处理。

有关其他信息，请参阅 [17]

6.5 Process section: Product realization / 过程阶段：产品实现

6.5.1 Design specifications/设计规范

During product development, the data from the Functional Specification(s) is transposed into design specifications and designs.

在产品开发期间，功能规范中的数据资料转化成设计规范和设计。

A **design specification** contains detailed descriptions of the hardware and/or software (including algorithms), all limitations and measurable values, including threshold values, the measures resulting from the Risk Analysis, all internal and external interfaces including clock times, and all measures used to prevent unintentional or deliberate incorrect operation.

Design Specifications are necessary when complex functions are realized. It is not necessary to generate an independent Design Specification as long as the function and interfaces are clear from a drawing or circuit diagram, and do not need additional information.

6.5.2 Realization of the Design/设计实现

The design is realized by transposing the requirements into the actual product. This includes generating drawings, parts lists, circuit diagrams (CAD designs), product-accompanying documents, as well as writing codes and generating samples.

The corresponding design measures from the Risk Analysis, from the standard and laws in agreement with the corresponding inspection station, as well as the design measures for environmentally-friendly product development must be observed.

6.5.3 Design verification measures/设计验证措施

Design verification measures are:

- Partial reviews (refer to chapter 6.7.4);
- Tests up to and including the system integration test.

From the stage of system integration onwards, Customer Service is responsible for maintaining proof of the install-ability (installation and startup) and serviceability of the product. Complete proof of install-ability must be provided in the system test at the latest. Proof of serviceability can be completed as part of the product introduction.

During the planning and performance of endurance tests, customer workflows and worst-case conditions must be considered.

The results of the design verification must be documented. Proof is provided via the corresponding test and review protocols. This is also applied to purchased hardware and software.

The test protocols prove that the guidelines from the test specifications have been observed, and that the expected results have been achieved. They summarize the results of a test run, and show whether the test has been completely and successfully performed. If problems/errors occur during the test, they must be listed and evaluated.

Simulations or calculations may be used to replace or supplement tests.

设计规范包括对硬件和/或软件(包括运算法则)的详细说明、所有公差范围和可测值,包括极限值,风险分析中提及的措施,所有内外联系接口包括时序,以及防止无意或故意的非正确性操作所采取的所有措施。

当功能复杂时,设计规范是必须的。只要图纸或线路图上的功能和联系接口是清楚明了的话,就不必单独生成设计规范,也不需要其它信息。

通过需求转化成实际产品,将设计变为现实。这包括生成图纸、备件清单、线路图(CAD 设计),随机文件以及编写代码和生成样机。

必须遵守风险分析中提出的相应设计措施,符合相应检验机构的法规以及为进行有利于环保的产品开发而采取的设计措施。

设计验证措施是:

- 部分评审(请参照 6.7.4 小节);
- 到系统集成测试为止并包括系统集成测试。

从系统集成开始,客户服务部门负责验证产品的可安装性(安装和启动)和可维修性。最终,必须在系统测试中提供完整的可安装性和可维修性证据。产品的可安装性和可维修性证据是产品发布的一部分。

在计划疲劳测试及其性能时,必须考虑客户的工作流程以及极端情况。

设计验证结果必须以文件形式写下来。通过相应的测试和评审报告提供证明。这同样适用于购买的软件和硬件。

测试报告证明测试规范中的指导方针已遵照实施,并且已经达到预期的目标。测试报告对各轮测试结果进行总结,并且表明测试是否已经全面并成功地进行。如果在测试期间出现问题/出错,必须列出来并进行评估。

通过模拟或使用计算方法可以用于取代或补充测试。

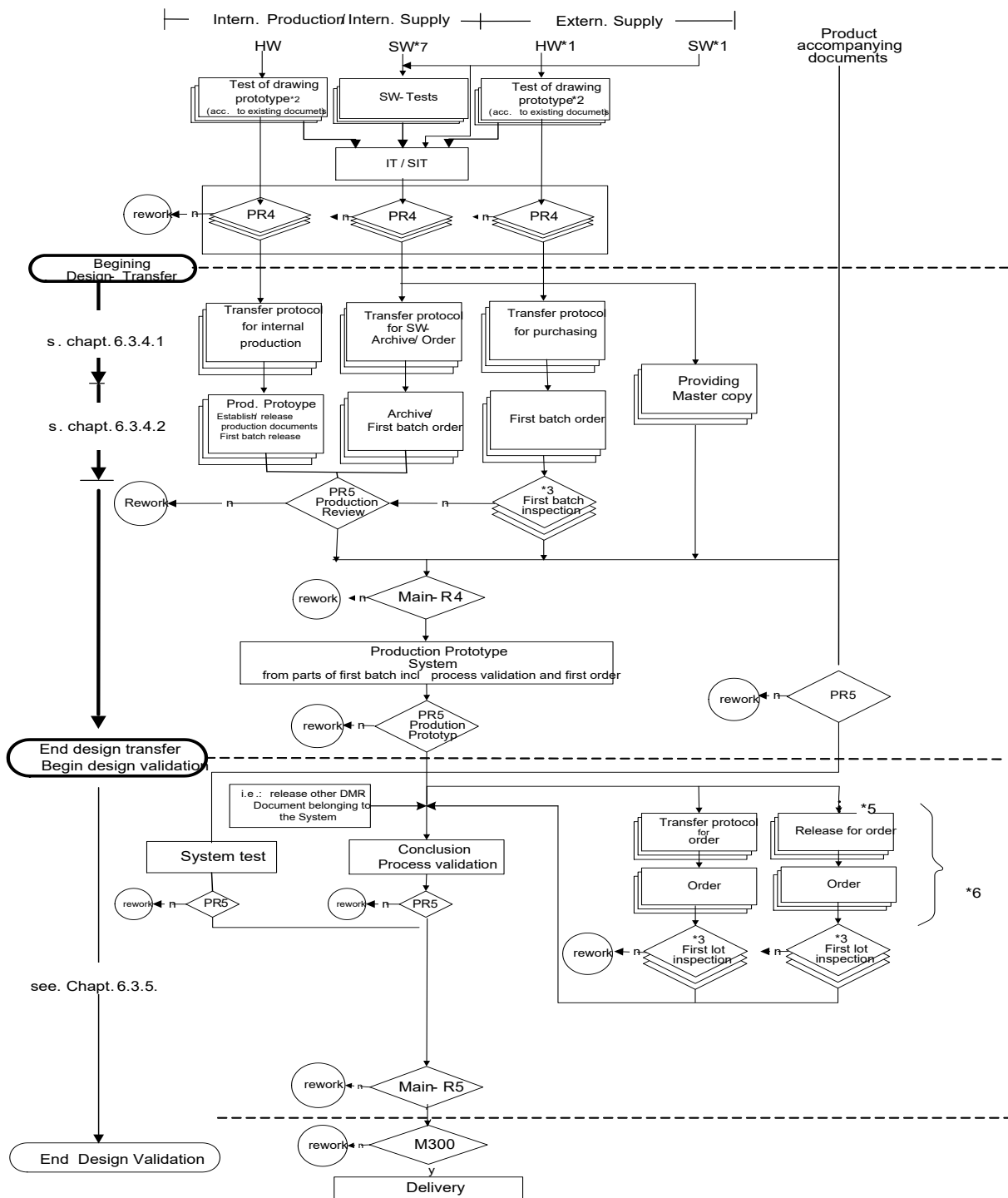
6.5.4 Design transfer /设计转移

The procedure for design transfer ensures that R&D results are systematically transferred to the production or the procurement site, so that the required initial lots for design validation can be manufactured. The necessary steps, as well as their requirements, are displayed in Fig. 6.

Concurrent Engineering supports the design transfer through the prompt involvement of production and procurement sites (purchasing and logistics), as well as vendors.

设计转移程序确保技术开发结果被系统地转移到生产或采购方面以便设计确认所需的首批产品可以进行生产。图 6 表明所需步骤以及需求。

同步设计通过生产部门和采购部门(采购和物流)及供货商的即时参与协助设计转移。



*1 The purchase process can be adapted to the specific requirements.

*2 with the agreement of QT, the functional sample can be replaced by calculation, review or simulation.

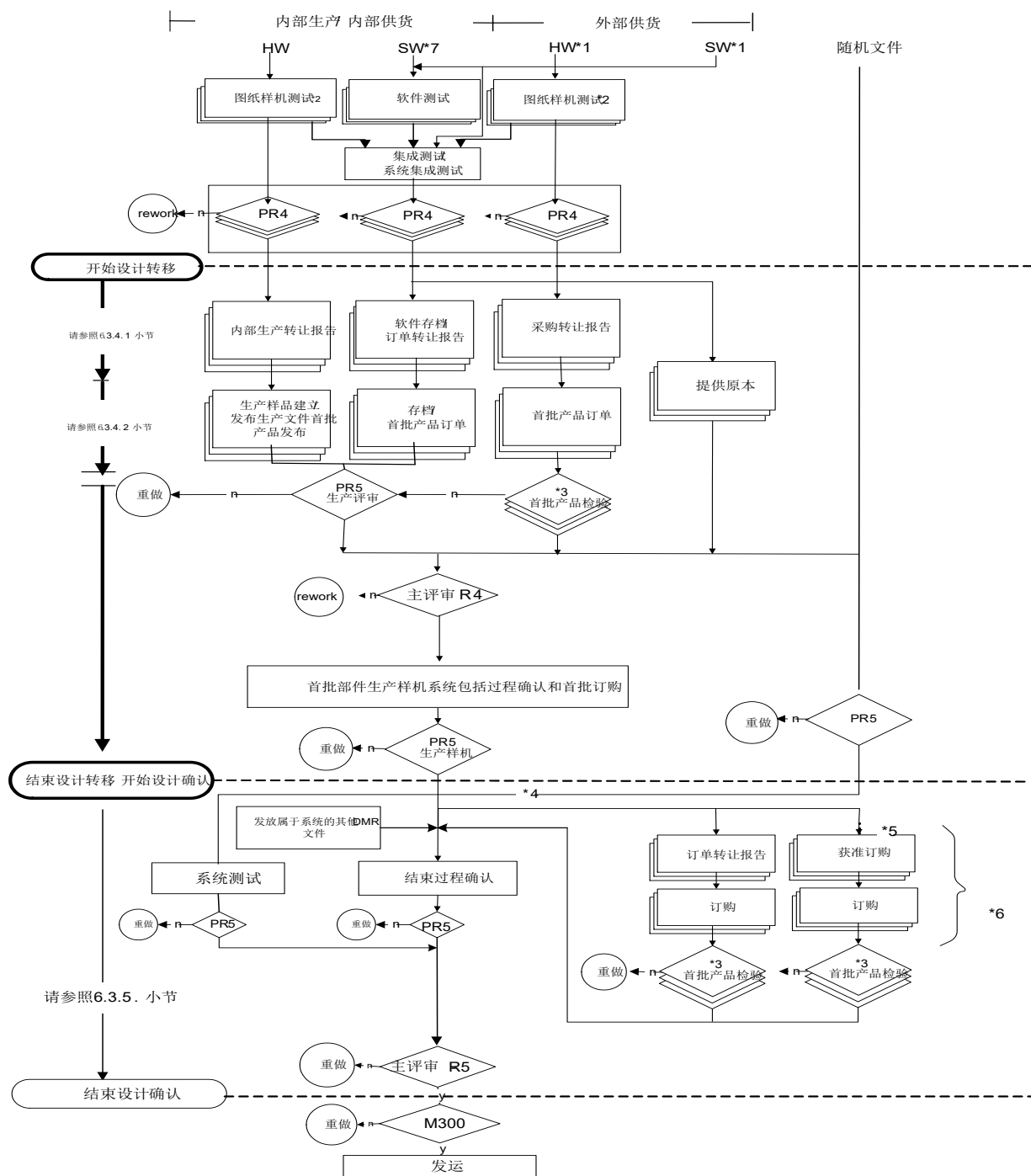
*3 The manner and scope of the first batch inspection should be determined according to the specific situation of the project.

*4 The product accompanying documents which are necessary for system test must be checked professionally and provided for the system test.

*5 Only for documents which are not validated during system test.

*6 Only for duplication of SW or printing order.

*7 Current SW version is according to Software Development Directive [3], only for system test (production prototype from first batch), this information should be included in SW Release Note.



- *1 采购过程可以适用于特殊的要求
- *2 QT 同意后，功能样品可以通过计算、评审或模拟来取代
- *3 根据项目实际情况，确定 首批检验的方式和范围。
- *4 必须专门检查系统测试必须的随机文件并且向系统测试提供这些文件。
- *5 只对系统测试期间没有确认过的文件
- *6 只对软件的复制或者打印的订单
- *7 此时的软件是按照软件开发指导 [3] 中规定的版本，仅为系统测试（首批部件生产样机）使用，该信息应在软件的“发布记录”中标明。

Fig. 6/图 6 Design Transfer and System Validation/设计转移和设计确认流程图案

Once the Function Review (partial review PR4) of a unit (e.g. assembly, component, sub-system) is completed, the design transfer begins, and the R&D results are transferred using the transfer protocol.

The transfer protocol:

Lists the entire volume of the documents to be transferred for the production of a unit, including document levels,

Formalizes the transfer of newly generated or documents that have been modified in the scope of an upgrade via PEP to production, serves as the release for the listed documents for initial lot production and making production tools available, for external purchases for release of the initial lot order, and distribute to SCM; if necessary, a copy is sent to the distribution list.

Electrical format of transfer protocol is also possible according to the written process of SAP.

The transfer protocol is generally signed by the author and the Process Engineer. The participation of additional departments, i.e. the QT or Customer Service, can be specifically determined, if necessary.

The signature confirms that:

- the contents of the planned transfer volume have been checked;
- the documents correspondingly coded in the transfer protocol have been received;
- the contents of the documents received have been checked (generally takes place before the transfer of the transfer protocol).

Changes to transferred documents with a completely signed transfer protocol are performed according to [1].

For changes to a completely signed transfer protocol, the following is valid:

For changes to the contents (i.e. supplementing or eliminating documents), the transfer protocol must be updated to the next version level and re-signed.

If Concurrent Engineering requires it, components may be submitted by R&D at different times.

A transfer protocol is then necessary for every individually transferred component.

If, due to dispositive reasons, it is necessary to order/manufacture parts before the function review (PR4), in these special cases, the parts from the initial lot can be used if:

一旦完成单元(如组件、部件、子系统)的功能评审(部分评审 PR4), 就开始设计转移, 并且用转移报告移交技术开发成果。

转移报告:

列出生产单元的全套文件包括文件级别, 表明产品设计过程中产品升级范围内最新生成的或修改的文件正式转移到生产中去。

用作所列文件已发放的证明, 可进行首批生产并可以使用生产工具, 可以进行外部采购, 可以发放首批订单, 转移报告应交给生产部门; 如果必要, 将复印件发给分发清单上的人员。

根据制订的相关 SAP 流程, 转移报告也可采用电子模板。

转移报告一般由作者和工艺工程师签名。如果需要, 其他部门(如质量或 客户服务部)也可以参与。

移交报告上的签名可以确认以下几点:

- 已经检查过计划转移的内容;
- 已经收到在转移报告中对应编号的文件;
- 已经检查过所收到的文件内容(一般在转移报告移交之前来做)。

转移报告经签字后, 这些被移交的文件上所做的修改应该根据参考文档[1]来做。

转移报告经签字后所做的修改, 下述情况是适用的:

内容上的修改(也就是文件补充或删除), 转移报告必须更新, 到下一个版本级别为止, 并重新签上名字。

如果同步设计需要转移报告, R&D 部门可以在不同时期提交单元。

然后, 给各部件出具各自的转移报告。

如果出于决定性的原因, 有必要在功能评审(PR4)之前订购/生产部件, 在这些特殊的情况下, 如果出现以下情况, 可以使用首批产品的部件:

- The transfer protocol contains a notation that the Function review (PR4) has not yet been performed. This must be counter-signed by QT. As in the typical case, the production process is validated during initial lot production. Once the parts are available, the function review must be performed and the notation should be removed from the transfer protocol. This proof must be available by the time of the system production review, at the latest.
- For the system test, a software version is used which has been exactly defined in the SW Configuration Management plan according to Software Development Directive [3], and has been explicitly released for the system test (master copy).
- 转移报告包括表明功能评审(PR4)还没有进行的标示。质量部必须签字。在典型情况下, 生产过程的确认为首批产品生产期间完成。一旦完成, 必须进行功能评审并且在转移报告中去除这个标示。上述工作必须最迟在系统生产评审之前完成。
- 对于系统测试, 应使用根据软件开发指导[3]建立的软件配置管理计划中明确规定的版本, 并且已经明确发布为系统测试(母盘)之用。

A template of Transfer Protocol is located in the Appendix 3.

移交报告样例在附件 3 中。

6.5.4.1 Start of production / initial lot production / initial lot order/开始生产/首批产品生产/首批订单

Based on the documents listed in the transfer protocol or product SW, any additional measures which are required for the start of production or initial lot order must be carried out by the responsible process, refer to table 1.

根据转移报告或产品软件上列出的文件, 生产开始或首批订单所需的任何其它措施必须由相应的流程实施, 请参照表 1。

Together with the initial lots the product-specific internal production processes are validated according to [4].

和最初运行记录一起, 产品相关的内部生产过程应根据[4]进行确认。

Production – type of unit 生产-单元类型	Example of measures 措施	Responsible process. 职能流程
Internal production - HW/SW 内部生产-软硬件	Generate or complete production and test documents, incl. test devices and test software. 生成或完成生产和测试文件包括测试器材和测试软件 Create production devices 制造生产器材 Process validation & verification 过程确认 Archive the master copy (SW) 母盘归档(软件)	Process planning 过程计划 Document center 文档中心
External production - HW/SW/ product-accompanying documentation 外部生产-硬件/软件/随机文件	Generate dispositive documents (i.e. DMR documents) 生成决定性文件(即 DMR 文件) Determine tests (external and/or internal) 确定测试(外部和/或内部) Send the transfer protocol or documents/data to Purchasing 将移交报告或文件/数据发往采购部 Retrieve the data and if necessary, order product SW/data 收回数据资料, 在必要时, 订购产品软件/数据资料 Vendor approval 供货商批准	Process planning 过程计划 Purchasing/采购 (also refer to [5]/也请参照[5])

Table 1/表 1

6.5.4.2 Production review/生产评审

The production review provides objective evidence of the process verification & validation (validation of the production process).

生产评审为过程验证和确认（生产过程的确认）提供了客观的依据。

The author of the production review is generally the Process Engineer responsible for the unit. The production review is released by the responsible Quality Engineer.

生产评审的作者一般是负责单元的工艺工程师。质量部负责人负责批准发放生产评审。

The signature on the production review protocol confirms that

在生产评审报告上签名，对以下内容进行认可：

- the unit fulfills its function according to the specifications;
- all of the DMR documents required for the production of the unit are released;
- all of the DHR documents required for production and testing of the initial lot are available.
- 部件的功能符合规范；
- 发放部件生产所需的所有 DMR 文件；
- 获得与首批产品生产和测试所需的所有 DHR 文件。

When the production review is released, the initial lot becomes the first series lot. At the same time, the approval for series production of the unit is granted.

Units from initial lot or series production may only be shipped if the M300 decision has been granted, except if the design validation is performed by the customer (refer to [chapter 6.5.5](#)).

A template of Production Review Protocol is located in the appendix 4.

6.5.5 Design validation / 设计确认

The design validation ensures that the product meets the user's specifications and requirements. It is performed at the end of the product development process phase, and contains the system test, as well as the acceptance test.

It must be ensured that every system is tested in its maximum or critical configurations, according to the system flow chart.

If products or product upgrades are validated at the customer site, it must be ensured that these non-released products / upgrades for patients cannot be used until they are released.

The validation of products is always performed at system level under simulated operating conditions.

During the validation, the user requirements (user needs and intended use) documented in the requirement specification must be tested.

All the other requirements (of internal and external stakeholders) from the requirement specification are tested during the design validation.

Design validation mainly comprises the following:

- Clinical evaluation to validate that the medical procedure is recognized, safe and effective;
- Usability to validate that the clinical workflow can be fully implemented;
- Reliability/system performance to validate that the system operates reliably and safely within the specified performance parameters.

当生产评审批准发放时，首批产品成为第一批系列产品。同时，同意批准进行单元的系列生产。

首批生产或系列生产的单元只有等到 M300 决定同意后才可发运，除非设计确认是由用户做的(请参照 [6.5.5 小节](#))。

生产评审报告样例在附件 4 中。

通过设计确认可以确保产品满足用户的预定规范和需求。应在产品开发这一过程阶段的最后进行设计确认，它包括系统测试以及验收测试。

必须根据系统流程图的要求，在其最大或最严格的配置条件下，对每台系统进行测试，

如果产品或产品升级是在用户现场进行确认，那么必须确保产品/产品升级在发布之前不能用于病人。

产品的确认总是在系统层基于模拟操作环境进行的。

在确认过程中，定义在需求规范文档中的用户需求（用户需要和设计应用）必须测试。

所有在需求规范中由内部和外部参与者定义的其他需求应在设计确认中测试。

设计确认包含以下内容：

- 通过临床评估来证明产品符合医疗程序，并且安全有效；
- 可用性确认可以实现临床流程
- 可靠性和系统性能用以确认系统可在特定环境下可靠并且安全地运行。

The following methods can be used to validate that the requirements have been implemented effectively/into the design and that the user needs have been met:

- Clinical evaluation => Clinical investigation or literature review;
- Usability => Workflow tests that represent the complete clinical workflow, performed by application personnel;
- System performance => Climate test, vibration test, packaging test, etc.;
- System availability/reliability => Endurance and stress tests.

Check documents which are published for customers, i.e. data sheets, Price Book, operating instructions, planning guide, confirms that the product characteristics have been met.

Confirm the operating instructions are appropriate for the intended use of the product.

If material that does not yet have a series number, Test Center requests a confirmation with a statement of equivalence to the serial material by the supplier.

可用以下方法来确认需求是否被实现，以及用户需要是否达到：

- 临床评估 => 临床试验或文献评审；
- 可用性 => 临床应用人员进行临床工作流程的测试；
- 系统性能 => 环境测试，振动测试，包装测试等；
- 系统可用度/可靠性 => 耐久和疲劳测试。

检查并确认在系统参数说明书，价格目录，操作说明书，机房设计指导等文档中，产品特性是否符合。

确认操作说明书对产品预期用途的描述是否正确。

如果系统测试使用的样品不是生产样品（没有序列号），测试中心将要求供应商提供一个和生产产品等同的声明。

6.5.6 Country-specific approvals / 各国市场准入 (各个国家的许可证制度)

Country-specific approvals can be linked to product-specific, as well as other normative requirements, with respect to the quality management system and its proven effectiveness.

The proofs and documentation necessary for acquiring approvals from countries of business are generated during the product development / production start-up process phase, in conjunction with the corresponding inspection office.

The application for approval is done in cooperation between sales, the respective subsidiary and the SSME QT.

In review R5 it has to be checked which of the required country-specific approvals are present.

各国市场准入关系到产品的专用需求，以及与质量管理体系和其效力有关的法规需求。

应在产品开发/生产启动过程阶段中，与有关检验机构合作，完成必要的证明和文件资料以申请经销售国家的市场准入。

许可证申请工作由销售部门、当地分支机构以及 SSME QT 部门共同合作完成。

在进行 R5 评审时就要检查已经有了所要求的国家的许可证。

6.5.7 Customer use test / 客户应用测试

A customer use test can be performed to obtain non-medical findings (e.g. handling, design acceptance, etc.) in a clinical environment. For this, the required country-specific releases and approvals must be available for the product. The QMP defines the number of products. Moreover, the duration of the customer use test and the criteria for selecting the customers who will be taking part in the customer use test are determined, refer to [10].

客户应用测试用来在临床环境中发现非医疗相关的问题。客户应用测试必须在获得相应国家的市场准入许可后开始。在 QMP 中定义进行客户应用测试的产品数量和周期，以及挑选客户的准则。具体的要求，参考[10]。

6.6 Process section: Product marketing and product phase-out /过程阶段: 产品市场化和逐步结束

6.6.1 Decision M300 /决定 M300

Product final release requires a M300 decision. Along with approval to ship the product, the service parts listed in the document “service parts catalogue” are also ready in stock for shipment.

Country-specific approvals

The delivery release contains approval for at least one country.

Countries, in which delivery is intended, but not yet permitted, the delivery block must be done in SAP.

If an amendment to the country-specific approvals is necessary, or the countries implement additional prerequisites, the new requirements must be incorporated via the Change Control Procedure, refer to [1].

Activities of product support

The following processes are important activities during product support:

- Manufacture (procurement), shipment (incl. generating the DHR), installation and start-up, generally during the order processing process
- Making repair and maintenance available, incl. service parts supply, generally during the service process
- Monitoring of product quality (i.e. complaint handling), generally during the product support process
- Error correction, improvements of the product and production process, acc. to the change procedure and in general during the product support process

They are a part of the respectively named process, yet they are not a part of this QR, and must be fulfilled according to the corresponding guidelines and the SSME XP specific determinations for this process.

6.6.2 End of delivery (M400)/ 终止产品发运 (M400)

With the decision M400, discontinuation of delivery, the stock levels and, where applicable, the remaining product support periods are determined.

When the last unit of a product is shipped, this marks the beginning of the period for additional service parts delivery requirements. Notification on product discontinuation to notified body is necessary in order to guarantee that the monitoring contracts with e.g. UL (USA) and CSA (Canada) are terminated to reduce the associated cost.

产品最终发布前需取得 M300 决定。产品被批准出厂的同时,“备件清单”文件中列出的所有备件已经有仓库备货准备发运。

各国市场准入许可

发货批准包括有至少一个国家的市场准入许可。

对准备发运、但还未取得市场准入许可的国家,必须在 SAP 中作发货禁止。

如果必须对特定国家的市场准入进行修改,或某些国家实行了另外的规定,必须按照修改控制程序(参考[1])来引入这些新的需求。

产品支持活动

以下过程是产品支持中的重要行为:

- 生产(采购),发运(包括生成 DHR),安装和启动,一般是在合同执行过程中发生;
- 维修保养的准备工作,包括备件供应,一般是在售后服务过程中发生;
- 产品质量监督(如:用户抱怨处理),一般是在产品支持过程中发生;
- 错误纠正,产品和生产过程的改进工作,一般是根据修改程序在产品支持过程中发生。

这些行为是各个过程的组成部分,但它们不是本 QR 的组成部分,必须根据相应的准则以及 SSME XP 对这一过程的特别规定来完成。

If product delivery is restored after having been discontinued, all the delivery requirements need to be checked, restored if necessary and/or the R5 main review stage needs to be replaced. A new M300 decision on release of delivery has to be made.

Detail refer to [12]

A template of M400 is located in the Appendix 15.

发运终止后，如果重新发运，必须检查或重新建立(如果需要)所有的发货要求。需要重新进行 R5 评审并且需要重新做 M300 决定发货。

具体请参考[12].

M400 样例在附件 15 中。

6.6.3 End of Support (M450) / 终止产品支持 (M450)

The M450 milestone makes a decision of end date for obligation to deliver spare parts (end of product support). Detail refer to [12]

A template of M450 is located in the Appendix 11.

M450 决定结束配件供应的日期（结束产品的支持）。具体参考[12]

M450 样例在附件 11 中。

6.6.4 M500 decision / M500 决定

At the end of the product lifecycle, it is decided when the affected product is no longer used. Milestone M500 is reached when all systems of the affected product are marked "scrapped" in the Installed Product Base tool.

The M500 decision is applied for by means of a project request by the head of the PSG. This decision regarding the end of the product lifecycle is made by the responsible R&D manager and the head of quality management in the XP BL. After the M500 request has been approved, the responsible PSG determines the required activities to end the "Product marketing and product phase-out" process section.

A template of M500 is located in the Appendix 12.

在产品生命周期结束时，需要决定不再使用受影响的产品的时间。当逐步退出的产品在现场装机数据库中都被标注为“scrapped”状态时，产品生命周期随之终止。（M500）

M500 的决定由项目 PSG 负责人提出要求。最终由研发总监和质量管理总监批准。决定批准后，相应的 PSG 决定所需的活动来保证“产品市场化和逐步结束“过程的结束。

M500 样例在附件 12 中。

6.7 Project handling / 项目处理

If a process organization is determined by SSME XP, the following tasks and activities must be transferred to the named business process /person responsible for the sub-process [9]. This also includes the tasks necessary for controlling and the continuous improvement of the product engineering process.

如果由 SSME XP 确定过程组织结构，以下的工作和活动必须移交给负责子过程所指定的经营过程/人员[9]。这也包括控制和继续改进产品设计过程的工作。

6.7.1 Task and responsibilities / 任务和职责

GM (Head) of SSME XP / SSME XP 总经理

- ensures that the project-specific tasks, responsibility, and authority of the project manager are determined and documented
- Makes decisions regarding project requests, based on documents provided by the project manager
- 确保确定并写下项目相关工作、项目经理的职责和权限；
- 根据项目经理提供的文件，对项目申请作出决定；

- formally agrees to the request with his signature. Approval for the required resources (investments, personnel and financial resources) is granted with this signature
- Shall establish an independent quality management representative
- Sign the CE Declaration of Conformity
- 签名表示正式同意项目申请。批准项目所需的资源，并签名表示同意(投资、人事和财务资源)；
- 确保建立一个独立的质量管理代表；
- 在 CE 符合声明上签字。

Quality management responsible person / 质量管理负责人

- for project request, confirms the agreement between the quality plan and the valid QM system;
- ensures that all regulations and laws valid for the project are followed;
- shall establish an independent quality engineer for each project/product;
- signs the CE Declaration of Conformity.
- 对于项目申请，确认质量计划与生效的质量管理体系的一致性；
- 确保项目遵守所有生效的法律和法规；
- 确保为每一个项目和产品有一个独立的质量工程师；
- 在 CE 符合声明上签字。

Project manager / 项目经理

- Is responsible for achieving the defined project objective within the budget and timeframe with complete satisfaction of the required scope of performance and quality.
- Proposes the PSG members in coordination with the particular functional areas; the PSG members are then appointed by the management of the SSME XP.
- Leads the project team and the PSG and is the representative of the project in its entirety to the management of the SSME XP (providing, for example, periodic project status reports and promptly reporting relevant project risks and plan deviations that cannot be regulated).
- Designates the members of the product introduction team in coordination with the respective functional areas.
- Ensures that, depending on the tasks and process section, the PSG is composed of experts from the functional areas, such as PRM, R&D, SCM, Customer Service, the commercial department, quality management and procurement.
- Authorized to issue instructions and to give work assignments to project members, but is not the disciplinary manager
- Plans and communicates the project objectives to project staff.
This includes, for example:
 - Communication of the project objectives and content to project staff
- 负责在预算和时间周期内完成项目的既定目标，并且达到需要的产品性能和质量要求。
- 与各职能团队协商，提议 PSG 成员名单。PSG 成员由 SSME XP 管理团队指定。
- 领导项目团队和 PSG，并在面对 SSME XP 管理团队时，作为项目团队的代表（比如，定期汇报项目状态，及时汇报项目风险，以及项目计划执行偏差）。
- 与各职能团队协商，指派产品发布团队成员。
- 按照不同的工作任务和过程阶段，确保项目管理小组由各职能部门的专家组成，如：产品管理，研发，供应链管理，客户服务，商务，质量管理和采购。
- 作为非人员经理，被授权可以解释和分配项目成员的工作内容。
- 计划并负责与项目成员沟通项目目标。
包括：
 - 与项目成员沟通项目目标和内容

- Efficient organization of work in the team. The project manager strives to strengthen and promote the motivation of team members
- Creation of overall structure plan
- Creation of an overall activity plan and division of tasks for the project team
- Contract award to outside departments/organizations
- Leading role in the creation of project-related reports, documents and final reports
- Preparing and updating the QMP/PMP, as well as document plans
- Information of the project steering group about the content of the QMP
- Ensuring the project documentation is filed in the DHF and DMR
- Ensuring the review and approval of technically critical decisions
- Providing support for patent applications

- 组织团队有效地工作。增强和提升团队成员的积极性
- 创建整体项目计划
- 创建整体项目活动计划并且为项目团队拆分任务
- 与外部部门和组织沟通
- 主导创建与项目相关的汇报，文档和最终报告。
- 准备和更新 QMP/PMP，以及文档计划
- 与 PSG 成员沟通 QMP 的内容
- 确保项目文件存档在 DHF 和 DMR 中
- 确保对于关键技术决定的审评和批准
- 协助专利申请

Quality Engineer/ 质量工程师

- The responsibility of the quality engineer is to control the assigned project as an independent controlling instance. The quality engineer is obliged to stop processes and projects when corresponding quality requirements are not fulfilled (e.g. product characteristics, intended use, compliance with QM system). The quality engineer escalates directly to the quality management representative of the SSME XP management.
- 作为独立的控制角色，质量工程师的职责是来控制批准的项目。如果相关的质量需求没有满足（例如，产品的特性，预期用途，与质量管理体系冲突等等），质量工程师将被迫停止项目和流程。质量工程师直接向 SSME XP 的质量管理负责人汇报。

Project team member / 项目团队成员

- Independent performance of the assigned work packages / activities
- Ontime completion with technical requirements satisfied
- Ensuring the quality of one's own work results
- Documentation of the work results
- Feedback on work progress to the project manager
- Constructive collaboration for quick resolution of technical challenges in the team
- Compliance with the planned effort for assigned work packages
- 独立完成所分配到的工作包 / 任务
- 按时提供满足技术需求的方案
- 确保自身交付工作的质量
- 存档工作内容
- 向项目经理反馈工作进度
- 提出建设性意见，快速寻求技术挑战的解决方案
- 按计划完成所分配到的工作包

Project steering group / 项目领导小组

- Project steering group (PSG) is an interdisciplinary work group made up, for example, of PRM, R&D, SCM, Procurement, Quality management, commercial department and XP Customer Service (CS project manager)
 - with a common objective
 - with strong reciprocal relationships
 - with a feeling of community (team spirit)
 - with common rules of play
- PSG is part of the project team
- PSG supports the project manager. The project manager assumes the responsibility of representing the interests of those functional areas that are not represented in the PSG
- PSG remains responsible for the product even after delivery, regardless of whether members of the group change
- When changes are made to the product requirement specification or functional specification, decides whether a new M150 or M200 decision is required
- After the M300 decision, the PSG is responsible for product monitoring and taking appropriate measures (CAPA)
- Each member of the PSG
 - Represents the functional areas which they were delegated
 - Assumes coordination and controlling tasks between the PSG and their functional areas
 - Makes project-related decisions in their functional areas
- The PSG meeting mins documents action items and summarizes decisions and latest information from the PSG meeting, and the topics in the PSG meeting mins is not mandatory based on project phase. If any of them is not applicable, N.A should be recorded for relevant topic. The example of the PSG meeting mins is shown in Appendix 14 for reference.

Product introduction team / 产品发布团队

- Is responsible for creating and implementing the product introduction plan
- Monitors and coordinates all activities in accordance with the project-specific product introduction plan with the aim of finding solutions to recognized problems

- 项目领导小组（PSG）是一个跨职能团队，通常由产品管理，研发，供应链管理，采购，质量管理，商务销售部门和 XP 客户服务组成（CS 项目经理），另外项目领导团队
 - 拥有共同的目标
 - 具有有牢固的相互关系
 - 富有团队精神
 - 执行同样的规则
- PSG 是项目团队的一部分
- PSG 帮助项目经理进行项目管理。项目经理承担代表未参加 PSG 的那些职能部门利益的责任
- PSG 在产品开始交付时仍然对产品负责，无论 PSG 成员是否变化
- 当产品需求规范或功能规范变化时，决定是否需要新的 M150 或 M200 决定
- 在 M300 后，PSG 负责产品的监控并实施相应措施（CAPA）
- 每个 PSG 成员
 - 是各个职能部门的代表
 - 协调功能部门和 PSG 之前的任务
 - 代表其所在的功能部门作与项目相关的决定
- PSG 会议纪要记录了行动项目，并总结了 PSG 会议的决定和最新信息。在 PSG 会议纪要中的主题是非强制性的，基于项目阶段填写，如果其中一个不适用，则记录不适用。PGS 会议纪要模板参考附录 14。

- 负责创建和实施产品发布计划
- 协调监控与项目相关的产品发布计划，并解决计划执行过程中发现的问题

6.7.2 Project requests/项目申请

- Are generally submitted by the Project Manager or PSG. This can be specifically arranged;
- Serve as documentation for the documents presented as the decision basis;
- Document the approval of the management at the start of the respective process phase.

Guideline for the change derived project:

Project is recommended for big changes which impacted to the approved country license with big cost and long timeline. Otherwise, it can be handled by change process. Decision is done by the XP Management team and recorded in QMP.

Examples for a project recommended as following:

- If new features either in HW or SW has been introduced, which need update the regulation license, a project is recommended
- If the change cost >100kEUR, a project is recommended
- If the change timeline >1 year, a project is recommended

- 一般由项目经理或项目管理小组提出。这些要求可以作出专门的安排。
- 可将其作为决策依据文件
- 可作为经营管理机构对启动各过程阶段的批准文件

对因为变更引起的项目的指导原则:

如果变更很大(大成本, 长时间)或者影响到相应国家的重新注册, 应该启动项目流程。否则, 变更流程可以适用。最终决定由 XP 管理团队决定并在 QMP 中记录。

以下是相应的例子:

- 如果硬件或软件的新特性需要在相应国家重新注册, 应该启动项目流程
- 如果变更成本大于 100K 欧元, 应该启动项目流程
- 如果变更时间大于一年, 应该启动项目流程

6.7.3 Project documents which are not quality-relevant/与质量无关的项目文件

They are for internal use only and shall not be made available to auditors/inspectors. They only contain documents required for proper business organization of the project.

This generally includes:

- The cost and accounting plan (condensed form of project resource planning);
- The product line plan;
- Target costing;
- The business product plan;
- The establishment of financial criteria for project termination;
- The project report for constant monitoring of costs, deadlines and results of goals;
- The project risk management plan (evaluation of risks from i.e. delayed processing of assignments, planning of counter-measures);
- The documents for risk management from a financial point of view (evaluation of the financial effects of project discontinuation, delay or increased use of resources);
- Profitability calculations for the make or buy concept;

这类文件仅供内部使用, 不提供给审计/检验方。这类文件仅仅是项目经营管理所需的文件。

这主要包括:

- 成本和财务计划 (是产品资源计划的紧凑格式);
- 生产线计划;
- 目标成本;
- 产品的经营计划;
- 制定项目终止的财务准则;
- 项目报告, 用于随时控制项目的成本目标, 时间目标和效果目标;
- 项目风险管理计划 (风险评估, 如由于某些工作延误所造成的风险, 风险防范措施计划);
- 从商务角度考虑而制定的风险管理文件 (对由于项目终止, 项目延迟或投入更多资源所造成的财务结果进行评估);
- 对自制或购买方案分别进行利润率计算;

- Process measurements (process measurement parameters such as actual process speed, use of resources, process cycle time, process quality (First Pass Yield), scheduling accuracy, process costs, efficiency, and extra costs due to follow-up work).
- 过程衡量（过程衡量参数，比如：实际过程速度，资源利用，过程周期，过程质量（一次通过率），时间计划准确性，过程费用，效率，由于后期工作所产生的额外费用）。

6.7.4 Reviews/评审

Main reviews serve to check that the documents created in a process section are complete, formally correct (technical review and approval) and consistent. The participants in the main review include representatives of all functions concerned with the stage of development under review as well as one person who has no direct responsibility for the stage of development under review and any specialists who also may possibly be required. The person who has no direct responsibility for the stage of development under review is the Q employee.

Partial reviews technically and formally test DHF and DMR documents, and are a prerequisite for their release.

The naming of review participants is generally done according to the QMP. With respect to the review participants, it should be ensured that not just the “main participants” (technical experts in generating documents) are taken into consideration, but also those who will later be affected by the contents of the document, i.e. those who are responsible for validation should participate in the review of the PRS.

主评审检查在流程部分定义的所需文件是否完整，正确（技术评审和批准）和一致性。主审查的参与者包括与所评审的发展阶段有关的所有职能的代表，以及对所评审的发展阶段没有直接责任的独立人员以及可能需要的任何专家。对被评审发展阶段没有直接责任的独立人员是质量工程师。

部分评审是对开发历史文档 DHF 和产品制造性文档 DMR 文件进行技术审核和形式审核，这是这些文件得以批准的前提。

一般来说，根据质量管理计划中的有关规定指定评审参加者。须明确的是，并不仅只“主要参加人员”（编制文件的技术专家），而且此文件内容将来涉及到的其他人员都应该参加评审，比如：负责确认的有关人员应该参加产品需求规范的评审。

6.7.5 Integration of AI algorithms / 集成 AI 算法

If AI algorithm is used as part of the medical device provided by SSME XP, related process must be fulfilled, refer to [16].

如果医疗设备产品包括 AI 算法，需要遵循相关的流程，参考[16]。

6.7.6 Change Control during Project Phase / 项目阶段的更改控制

All the changes related to products or documentation shall be controlled.

Before System/Subsystem R4, all the changes shall be followed up by the Defect Management process to document the difference, impact and the actions.

After System/Subsystem R4, all the changes shall be strictly performed according to [1].

在项目阶段，所有与产品和项目相关的变更必须被控制。

在系统和子系统 R4 之前，所有的变更必须按照缺陷管理流程实施，记录变化，影响以及应采取的措施。

在系统和子系统 R4 之后，所有的变更必须严格按照[1] 修改控制程序执行。

7. Provisional solution and backward method/过渡措施和补救办法

This procedure will be executed for all the SSME XP products and new projects starting from the effective date

For on-going projects before the main review R2, it is mandatory to follow this directive. If the project is between main review R2 and R5, the QMP must be supplemented specifying how and when a transition will be made to the PEP. The transition must be completed 6 months after the validity date of these instructions at the latest. Process phases which have already been concluded through the main review R5 do not need to be revised.

该文件在生效日起即对所有 SSME XP 产品和新项目生效。

对于正在进行的、主评审 R2 之前的项目，则必须执行此程序。如果项目介于 R2 和 R5 主评审之间必须在质量管理计划中补充说明如何以及何时进行 PEP 的过渡。最晚必须在本指令生效后的 6 个月内完成过渡。对于已经结束并通过主评审 R5 的过程阶段不必再进行修改。

8. Record/记录

Record no. 记录编号	Record name 记录名称	Where stored. 保存地	Retention 保存期限
n.a.	n.a.	n.a.	n.a.

9. Appendix/附录

Appendix no. 附录号	Appendix name 附录名称
1	Example of using the V-model in sub-systems 在子系统中使用 V 字型模式的实例
2	Example of “Project Decision Protocol” “项目决定报告”的实例
3	Example of “Transfer Protocol” “转移报告”实例
4	Example of “Production Review Protocol” “生产评审报告”实例
5	Example of DMR/DHF document index DMR/DHF 文档索引的实例
6	Template of R1
7	Template of R2
8	Template of R3
9	Template of R4
10	Template of R5
11	Template of M450 Decision Protocol
12	Template of M500 Decision Protocol
13	Template of DMR
14	Template of PSG meeting minutes
15	Template of M400 Decision Protocol

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